

"Study of finite abelian extensions of $\mathbb{Q}$"

Master's Thesis

submitted to

Indian Institute of Science Education and Research Tirupati

in partial fulfillment of the requirements for the

BS-MS Dual Degree Programme

by

Adithya S

Roll Number: 201701042

Supervisor: Prof. D.S. Nagaraj

Professor and Chair of Mathematics, IISER Tirupati

March, 2022

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Certificate

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Name of student: Adithya S

Roll Number: 201701042

Signature of student: Adithya. S.

Date: April 8, 2022

Name of Supervisor: Prof. D. S. Nagaraj

Name of Co-Supervisor:

Signature of Supervisor: Nagaraj.D.S.

Signature of Co-Supervisor:

Date: April 9,2022

Date:

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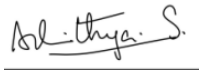
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Roll Number: 201701042

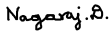
Signature of student: 

Date : April 8, 2022

Endorsed by:

Name of Supervisor: Prof. D. S. Nagaraj

Name of Co-Supervisor:

Signature of Supervisor: 

Signature of Co-Supervisor:

Date: April 9, 2022

Date:

Acknowledgments

I am very grateful to the faculty at IISER Tirupati who have been very supportive, kind and encouraging, especially during difficult times. I am blessed to have had the opportunity to work and study with/from the people in the mathematics department who have been the most kind, motivating and encouraging people I have come across.

I am thankful to my MS thesis supervisor Prof. D. S. Nagaraj, for their support, words of wisdom, encouragement and the discussions which I have immensely benefited from. I am also thankful to my mentor, Dr. Venketasubramanian C.G., for their constant support, motivation, and encouragement since my first day at Tirupati, and also for their boundless passion for mathematics and its pedagogy.

I am thankful to my dear friends and family for their unconditional love and support. I also thank Harshitha, Thamarai, Abhinaba, Lalith, Abhinav and the other members of the math club for fruitful discussions that I have benefited a lot from.

I finally wish to thank my teachers Prof. D. S. Nagaraj, Dr. Venketasubramanian C. G., Dr. Anilatmaja Aryasomayajula, Dr. Lakshmi Lavanya, Dr. K. P. Yogendran and all others who have inspired me to pursue a research career in mathematics.

Abstract

The main aim of this project is to understand the finite abelian extensions of the field of rational numbers $\mathbb{Q}$. This is basically a reading project. The study involves understanding various completions of number fields, which are finite extensions of $\mathbb{Q}$. This is the first part of the project. In the second part of the project, we wish to apply various results from the first part in order to complete the study of the finite abelian extensions of $\mathbb{Q}$.

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Chapter 0

Introduction

A Diophantine equation is a of polynomial equation in one or are more variables with coefficients in integers and seeking integer solutions. i.e., an equation of the type $f(X_1, \ldots, X_n) = 0$ where $f(X_1, \ldots, X_n) \in \mathbb{Z}[X_1, \ldots, X_n]$, for some $n \in \mathbb{N}$, and only integer solutions are allowed. Algebraic number theory involves the study of solutions to diophantine equations. The problem of finding solutions or showing the existence or non-existence of solutions to diophantine equations has remained at the heart of number theory for many centuries. The study of integer solutions to seemingly simple diophantine equations has been central to many fundamental breakthroughs in number theory. A classic example of such a diophantine equation is the Fermat's last conjecture (FLT)[1], a problem which had remained unsolved/unproven for nearly 358 years, asking for a proof that there does not exist positive integer solutions to the equation $x^n + y^n = z^n$, with $(x, y, z) \in \mathbb{N}^3$, and for $n \geq 3$.

On the other hand, after being fascinated with the laws of quadratic residues, Gauss gave more than 6 different proofs for what he called his *Theorema Aureum* (The Golden Theorem), which is the *Quadratic Reciprocity Law*. He later, upon investigating the arithmetic of the ring $\mathbb{Z}[i]$ in search for a proof of the law of bi-quadratic (a.k.a. quartic) reciprocity law and failing to find one, is said to have asserted that the proof belonged to the mysteries of the higher arithmetic[2]. It is precisely these generalized integer rings (known as *number rings*) such as $\mathbb{Z}[i]$ which fall under the broader category of Dedekind domains that we have started our study with (see Chapter 1). We could perhaps safely say that the search for higher reciprocity laws and the proof of Fermat's last conjecture had been one of the key

[1] Now a theorem due to the first correct proof in 1995 by Andrew Wiles

[2] See the introduction (p. 108) and §7 (p. 119) of Chapter 9 in [Ireland-Rosen]

driving forces for the development of modern number theory in the past two centuries.

In this project, we have taken up the study of algebraic number theory – involving the study of the theory of *ideal numbers* due to Kummer and Dedekind, the theory of number fields, their abelian extensions, and their invariants (– the different, discriminant, class group, etc.), the geometry of numbers (Hermite–Minkowski theorem), and the structure of number rings (Dirichlet's unit theorem) and their connections to diophantine equations. We have also studied the Kronecker-Weber theorem, which says that cyclotomic extensions capture all the finite abelian extensions of $\mathbb{Q}$. We have concluded our study with the theory of *class fields*[3], which gives a description of the abelian extensions of global and local fields. We have studied global class field theory from [Janusz], which follows an analytic approach for some of the mains results and doesn't involve the need of heavy machinery such as the cohomology of groups in [CasFrö].

While Kummer's work on the theory of ideal numbers led to the proof of a special case of the FLT for an infinite class of prime numbers in 1847 (see Section 1.1 of Chapter 1), the work of Gauss, Eisenstein, Kummer, Hilbert, Tagaki, Hasse, Artin and others on class field theory led to a general reciprocity law, namely the Artin reciprocity theorem as a special case[4].

We briefly outline the contents of this report below.

In Chapter 1, we discuss the prerequisites from commutative algebra such as the theory of Dedekind domains, discrete valuations rings (DVRs), the notion of module index, the RKLS-Setup and its invariants – which is our main focus of study, and the concept of ramification. We shall also assume some background in basic Galois theory such as that of Chapters 13 and 14 of [Dummit-Foote].

In Chapter 2, we outline the proof of the Kronecker-Weber theorem using ramification theory and geometry of numbers. We also give a proof of the inverse Galois problem for finite abelian groups.

In Chapter 3, we discuss some results about number fields, such as the existence of an integral basis, the structure of the number ring as given by Dirichlet's unit theorem, the ideal class groups, and we conclude with a method to compute the same in some special

[3]The term *"class field"* refers to a field extension (of a number field) satisfying a technical property that is historically related to ideal class groups, and one of the main results is that class fields are the same as abelian extensions. For more details, see K. Conrad's marvelous article [Conrad-CFT-History].

[4]See [Lemmermeyer] for a detailed account of the history, evolution and a discussion of the modern state of the reciprocity laws.

cases.

In Chapter 4, we discuss some results from class field theory. Our main results in this chapter are about the applications of Dirichlet density and some analytic methods to prove the Frobenius density theorem, which in turn is used to get the surjectivity of the Artin map and prove the existence of the Hilbert class field of any number field.

In Chapter 5, we discuss some examples and list our computations. In this chapter, we have used the results from Chapters 1-4 of this report, discussed some new ideas and methods to find integral bases and/or compute the class groups of some quadratic and cubic (abelian) extensions of $\mathbb{Q}$. We have used the SageMath software (see [Sage] and [Stein] for more details) for some of our computations.

Chapter 1

Preliminaries

1.1 Theory of Dedekind Domains and DVRs

We first discuss the theory of Dedekind domains and Discrete valuation rings (DVRs), for which we refer to [[CasFrö], Chapter I, Sections 1 and 2] and [[Atiyah-MacDonald], Chapter 9]. A discrete valuation ring (DVR) is a local domain (i.e., a local ring which is also an integral domain) whose non-zero prime ideal is a maximal and a principal ideal. Dedekind domains are *locally* (i.e., when localized at any of their non-zero prime ideals) DVRs.

Dedekind domains arise naturally as rings of integers of number fields, as we shall see later (See Example 1.1.5). We first discuss a few interesting results about Dedekind domains.

For any integral domain R and its field of fractions $K = \mathrm{Frac}(R)$, we define a *fractional ideal* I of R to be an $R-$submodule of K such that $I \neq (0)$ and $\exists\, \alpha \in K^\times$ such that $\alpha I \subset R$. Whenever this integral domain R is Noetherian, fractional ideals of R are precisely finitely generated $R-$submodules of K. Let $K^\times$ denote the multiplicative group of a field K, and let $\mathbb{Z}$ be the group of integers under addition. We define the notion of a discrete valuation v on K as follows:

Definition 1.1.1 (Discrete Valuation). A discrete valuation v on K is a map $v : K \to \mathbb{Z} \cup \{\infty\}$ such that the following three properties are satisfied:

1. v is a surjective group homomorphism from $K^\times$ to $\mathbb{Z}$. i.e., $v(xy) = v(x) + v(y)$.
2. $v(x + y) \geq \inf\{v(x), v(y)\}$.
3. We define $v(0) := \infty$, where the symbol ∞ is such that $\infty > x$, for all $x \in \mathbb{Z}$.

We note that when v is a discrete valuation on K, the *valuation ring* R_v of K defined to be $R_v := \{x \in K \mid v(x) \geq 0\}$ is a DVR. For a valuation ring R_v, we define the *valuation*

ideal $\mathfrak{p}_v := \{x \in K \mid v(x) \geq 1\}$, which is a principal ideal generated by an element $\pi \in R_v$, called the *uniformizer*, which satisfies $v(\pi) = 1$.

We also have that any DVR R is the valuation ring R_v for a *unique* discrete valuation v on $K := \mathrm{Frac}(R_v)$, where, $\mathrm{Frac}(R)$ denotes the field of fractions of a ring R. We hence characterize DVRs as follows: An integral domain R is a DVR if and only if R is Noetherian, integrally closed, and possesses a unique non-zero prime ideal. Dedekind domains on the other hand are a generalized notions of DVRs. They are Noetherian, integrally closed and its non-zero prime ideals are maximal. (i.e. their Krull dimension is one.)

Dedekind domains are not unique factorization domains (UFDs) in general, but, if they are UFDs, they are principal ideal domains as well. The assumption that certain number rings (which come under the category of Dedekind domains, as remarked earlier) are UFDs was essential in the earlier proofs of the *Fermat's Last Theorem (FLT)* by Lamé (in 1847) and others. This assumption was indeed false, in general (for example, the number ring of $\mathbb{Q}(\zeta_{23})$), and led Dedekind, Kummer, and others to develop the theory of *ideals*[1] and extend the proof of FLT to the so-called[2] *regular primes*. (See [[Marcus], Chapter 1] for details). For a proof of FLT for regular primes, see [Conrad-FLT].

Nevertheless, in Dedekind domains, there is a **unique** factorization of all of its *fractional ideals* into a product of finitely many prime ideals. In addition, for any Dedekind domain R there is a multiplicative group structure on the set of all fractional ideals of R, which is abelian and free on the non-zero prime ideals of R. We thus have the following characterization of Dedekind domains (See [[CasFrö], Chapter I, Section 2, Proposition 1 (p. 6)]):

The following conditions on an integral domain R are equivalent.

1. R is Noetherian, integrally closed and its non-zero prime ideals are maximal.
2. R is Noetherian, and, for every $\mathfrak{p} \neq (0)$, $R_{\mathfrak{p}}$ is a DVR (Discrete Valuation Ring).
3. All fractional ideals of R are *invertible*. i.e., for any fractional ideal I of R, there exists a fractional ideal I^{-1} of R such that $I \cdot I^{-1} = R$.

An Integral domain R satisfying any of the equivalent conditions mentioned above is called a *Dedekind domain*.

We restate the above discussed results as follows:

[1] Kummer developed his theory of *ideal numbers*, which restores the unique factorization for ideals, if not for elements, in all number rings. He also investigated the number rings of cyclotomic fields which he called cyclotomic integers.

[2] For a prime number p, we call p regular when the class number $h_p = h(\mathbb{Q}(\zeta_p))$ of the pth cyclotomic field is not divisible by p.

PROPOSITION 1.1.2. The set of all fractional ideals of a Dedekind Domain R form an abelian Group $\mathcal{J}_K$ under multiplication. The group $\mathcal{J}_K$ is free on the non-zero prime ideals $\mathfrak{p}$ of R. Furthermore, the representation of a fractional ideal I in terms of these generators is given by

$$I = \prod_{\mathfrak{p}} \mathfrak{p}^{v_{\mathfrak{p}}(I)}$$

Proof. Chapter 1, Section 2, Proposition 2 (p. 8) of [CasFrö]. □

We now define the notion of *valuations*.

DEFINITION 1.1.3 (Non-Archimedean Absolute value / Valuation). A non-Archimedean absolute value $|.|$ on a field k is a map $|.|\colon k \to \mathbb{R}_{\geq 0}$ such that the following axioms are satisfied.

(1) $|a| = 0 \iff a = 0$, for all $a \in k$.

(2) $|ab| = |a|\,|b|$, for all $a, b \in k$.

(3) $|a + b| \leq \max\{|a|, |b|\}$, for all $a, b \in k$.

DEFINITION 1.1.4 (Archimedean Absolute value / Valuation). An Archimedean absolute value $|.|$ on a field k is a map $|.|\colon k \to \mathbb{R}_{\geq 0}$ such that axioms (1) and (2) in Definition 1.1.3 are satisfied along with (3') below, which is the triangle inequality. i.e.

$$(3') \quad |a + b| \leq |a| + |b|, \ \forall\, a, b \in k.$$

We note that for a real number $0 < c < 1$, a discrete valuation v on a field K induces a Non-Archimedean absolute value $|\ .\ |_v$ given by $|x| := \begin{cases} c^{v(x)} & \text{if } x \in K \\ 0 & \text{if } x = 0 \end{cases}$

$|\ .\ |_v$ now induces a metric $d_v : K \times K \to \mathbb{R}_{\geq 0}$ on K, given by $d_v(x, y) := |x - y|$.

Let R be a Dedekind domain and $\mathfrak{p}$ be a non-zero prime ideal of R. Let $v_{\mathfrak{p}}$ denote the discrete valuation on K corresponding to $\mathfrak{p}$. Let $\overline{R}_{\mathfrak{p}}$ denote the valuation ring of the completion $K_{\mathfrak{p}}$ of K with respect to the metric induced by the absolute value $|\ .\ |_v$. We shall call this completion $K_{\mathfrak{p}}$ of K as the completion of K at $v_{\mathfrak{p}}$ in the sequel.

EXAMPLE 1.1.5 ($\mathfrak{p}-$adic absolute value). Let K be a number field, i.e. a finite extension of $\mathbb{Q}$. An *algebraic integer* is a complex number which satisfies some monic irreducible polynomial over $\mathbb{Q}$ with integer coefficients. Let $\mathcal{O}_K$ denote the set of all algebraic integers in K. ($\mathcal{O}_K$ is known as the *ring of integers* or the *number ring* of K). As a result of [[Janusz], Chapter I, Section 6, Theorem 6.1 (p. 26)] we conclude that $\mathcal{O}_K$ is a Dedekind domain.

For any non-zero prime ideal $\mathfrak{p} \subset \mathcal{O}_K$, let $v_{\mathfrak{p}}$ denote the discrete valuation on K induced by the prime $\mathfrak{p}$. We now define a Non-Archimedean absolute value called the $\mathfrak{p}-$*adic*

valuation $|\cdot|_{\mathfrak{p}}$: $K \to \mathbb{Z} \cup \{\infty\}$ as follows: Fix a real number c such that $0 < c < 1$. Now, the $\mathfrak{p}-$adic absolute value $|\cdot|_{\mathfrak{p}}$: $K^{\times} \to \mathbb{R}$ is given by $|x|_{\mathfrak{p}} = c^{v_{\mathfrak{p}}(x)}$. We extend $|\cdot|_{\mathfrak{p}}$ to K be defining $|0|_{\mathfrak{p}} = 0$.

1.1.1 Module Index

We now discuss an important concept called the *module index* in the context of Dedekind domains. For proofs and other details, we refer to [[CasFrö], Chapter I, Section 3] and [[Sutherland], Lecture #6].

Let R denote a Dedekind Domain, K its field of fractions, and V, a finite dimensional vector space over K of dimension $n \in \mathbb{N}$. An *R-lattice M* in V is a finitely generated R-submodule of V which contains a K-basis for V.

We first define the module index $[M : N]_R$ of free R-lattices M and N in V. We then extend this definition to any two R-lattices in V. This concept of module index shall be used to describe some invariants associated to a number field such as, the discriminant or the different.

Assume first that M and N are free R-lattices in V. As M and N are of rank n (because they both contain a K-basis each for V), they are isomorphic as R-modules. i.e., $M \cong R^n \cong N$. Suppose that ϕ is an R-module isomorphism between M and N. By fixing a $K-$basis $\mathcal{B}_M$ in M and expressing a fixed $K-$basis $\mathcal{B}_N$ of N in terms of $\mathcal{B}_M$, we get a unique K-linear isomorphism $\hat{\phi} : V \to V$ defined by the action of ϕ on the $K-$basis $\mathcal{B}_M$. We now define the module index $[M : N]_R$ as the fractional ideal of R generated by $\det(\hat{\phi}) \in K^{\times}$. i.e.,

Definition 1.1.6 (Module index for free R-lattices in V). Assume that M and N are free R-lattices in V. Let $\hat{\phi}$ be the K-linear map from V to V which is obtained from the R-module isomorphism $\phi : M \to N$ as indicated above. Then, the module index

$$[M : N]_R := (\det(\hat{\phi})) = \det(\hat{\phi}) \cdot R \in \mathcal{J}_K.$$

Now, we extend Definition 1.1.6 to any two R-lattices in V, which are not necessarily free R-modules using the following lemma.

Lemma 1.1.7. Let M be any R-lattice in V. Then, its localisation $M_{\mathfrak{p}}$ at any prime $\mathfrak{p}$ of R is a free $R_{\mathfrak{p}}$-module.

Proof. We first note that $R_{\mathfrak{p}}$ is a DVR, and hence a PID (Principal Ideal Domain). Now, the localization $M_{\mathfrak{p}}$ of M at $\mathfrak{p}$ is a torsion-free $R_{\mathfrak{p}}$-module, as it lies in the $K-$vectorspace V.

And thus, from the structure theorem for finitely generated modules over PIDs, any finitely generated module over a PID is free. Hence, $M_{\mathfrak{p}}$ is a free $R_{\mathfrak{p}}$-module. □

Assume that M and N are any two R-lattices in V, (which are not necessarily free). But, from Lemma 1.1.7, $M_{\mathfrak{p}}$ and $N_{\mathfrak{p}}$ are free $R_{\mathfrak{p}}$-modules, and hence free $R_{\mathfrak{p}}$-lattices in V. Thus, the module index $[M_{\mathfrak{p}} : N_{\mathfrak{p}}]_{R_{\mathfrak{p}}} \in \mathcal{J}_{R_{\mathfrak{p}}}$ is well-defined (from Definition 1.1.6) as a fractional ideal of $R_{\mathfrak{p}}$. We also note that if $M_{\mathfrak{p}} = N_{\mathfrak{p}}$, then
$[M_{\mathfrak{p}} : N_{\mathfrak{p}}]_{R_{\mathfrak{p}}} = R_{\mathfrak{p}}$. Now, as $[M_{\mathfrak{p}} : N_{\mathfrak{p}}]_{R_{\mathfrak{p}}} = R_{\mathfrak{p}}$, for almost all primes $\mathfrak{p} \in R$, and as
$\mathcal{J}_K \cong \prod_{\mathfrak{p}} \mathcal{J}_{R_{\mathfrak{p}}}$ [see [CasFrö], Chapter 2], there exists a unique fractional ideal $[M : N]_R \in \mathcal{J}_K$ such that, for all non-zero prime ideals $\mathfrak{p} \subset R$,

$$[M : N]_R \cdot R_{\mathfrak{p}} = [M_{\mathfrak{p}} : N_{\mathfrak{p}}].$$

Hence, we have the following definition.

Definition 1.1.8 (Module index for R-lattices in V). Let M and N are any two R-lattices in V. The module index $[M : N] := [M : N]_R$ is defined as the unique fractional ideal of R such that $[M : N] \cdot R_{\mathfrak{p}} = [M_{\mathfrak{p}} : N_{\mathfrak{p}}]$, for all prime ideals $\mathfrak{p}$ in R and $[M_{\mathfrak{p}} : N_{\mathfrak{p}}]$ is defined as in Definition 1.1.6.

Remark 1.1.9. From Definition 1.1.8, it follows that, if M and N are any two R-lattices in V, then $[M : N]_R = \bigcap_{\mathfrak{p}} [M_{\mathfrak{p}} : N_{\mathfrak{p}}]_{R_{\mathfrak{p}}}$. We trivially have that $[M : M]_R = R$. We also have a few interesting properties of module indices, which are easy to prove locally. The global form then follows (see [[CasFrö], Chapter I, Proposition 1, p.10] for details). (The subscript R in $[M : N]_R$ is often dropped and written just as $[M : N]$ when the Dedekind domain R is fixed, or understood in context.) For any R-lattices M , N and L in V, we have that,

1. $[M : N] \cdot [N : L] = [M : L]$
2. The inverse of the fractional ideal $[M : N]$ in $\mathcal{J}_K$ is $[N : M]$.
3. If $M \subset N$, then $[N : M]$ is an integral ideal of R. Moreover, $[M : N] = R \Rightarrow M = N$.

Remark 1.1.10. We are most interested in the case when V is a number field or a finite abelian extension of a number field, i.e., $V = K$, where K is a finite extension of $\mathbb{Q} = \mathrm{Frac}(\mathbb{Z})$ or a finite abelian extension of a number field L. In the former case, as $\mathbb{Z}$ is a principal ideal domain, the fractional ideals of $\mathbb{Z}$ are all principal. Any $\mathbb{Z}-$lattice is a fractional ideal of $\mathbb{Z}$. Thus, the module index of any two $\mathbb{Z}-$lattices in K is a $\mathbb{Z}-$submodule of K generated by a rational number.

From Item 3 in Remark 1.1.9, in the case when $R = \mathbb{Z}$, and $V = K$ is a number field, for two $\mathbb{Z}-$lattices $M \subset N$ in K, the notion of module index generalizes the group index, as we obtain that the module index $[N : M]$ is just the group index and corresponds to the colon ideal $(M : N)$ (Note the change in the order of N and M). [See [Sutherland] Lecture #6, p. 3 for details].

Furthermore, $[N : M] = (n)$, for some integer $n \in \mathbb{Z}$. Furthermore, if we impose that $n \geq 0$, we get that $[N : M]$ is an ideal of $\mathbb{Z}$ generated by a *unique* positive integer. Furthermore, as a result of [[Janusz], Chapter 1, Section 8, Proposition 8.4, p. 45], this unique positive integer, say, n, is the number of elements in the ring N/M, i.e., $[N : M] = (|N/M| \cdot \mathbb{Z})$.

1.1.2 The RKLS-Setup

Let R be a Dedekind domain, K is its quotient field and L, a finite separable extension field of K. Let S be the *integral closure* of R in L. i.e., $S := \{x \in L \mid x \text{ is integral over } R\} = \{x \in L | f(x) = 0, \text{for some monic polynomial } f(X) \in R[X]\}$. This setup shall be known as the RKLS-*Setup*.

We note that if $\mathfrak{p}$ is a non-zero prime ideal of R, then, $S \cdot R_{\mathfrak{p}}$ is the integral closure of $R_{\mathfrak{p}}$ in L. We say that a prime ideal $\mathfrak{P}$ of S lies over the prime $\mathfrak{p}$ of R if $\mathfrak{P} \cap R = \mathfrak{p}$ and we write this as $\mathfrak{P} \mid \mathfrak{p}$. We then have that every prime $\mathfrak{P}$ of S lies over a prime $\mathfrak{p}$ of R, and for every prime $\mathfrak{p}$ of R, there is a prime of S lying over it. In fact, we get a surjective map $\phi : \text{Spec}(S) \to \text{Spec}(R)$ given by $\mathfrak{P} \mapsto \mathfrak{P} \cap R$. (See [[Sutherland], Lecture #5, Lemma 5.28, p. 8] for details.)

We also have the following proposition.

Proposition 1.1.11. Assume the RKLS-Setup. Further, if R is a DVR, and if K is complete, then S is a DVR and L is Complete.

Proof. Chapter II, Section 3, Theorem 3.3 (p. 103) [Janusz]. □

1.1.3 Discriminant and Different

We now define a non-degenerate symmetric bilinear form $B : L \times L \to K$ on a number field L, which is a finite extension of a number field K.

Let for $x \in L$, define the *(multiplication by x)* K-linear map $m_x : L \to L$ given by $m_x(y) := x \cdot y$. Let the trace map $T = T_{L/K} : L \to K$ be defined as the trace of the K-linear

map m_x. i.e., $T_{L/K}(x) := trace(m_x) \in K$.

Let $B(u, v) := T_{L/K}(u \cdot v), \forall\, u, v \in L$. This defines a non-degenerate symmetric bilinear form on L (see [[Sutherland], Lecture #5, Theorem 5.20, p. 6] for details).

We also define the norm map $N = N_{L/K} : L \to K$ as $N_{L/K}(x) := \det(m_x) \in K$.

Definition 1.1.12 (Dual basis). Let L/K be a finite extension of number fields. Let $B : L \times L \to K$ be a non-degenerate, symmetric K-Bilinear form on L. If $\{u_i\}_{i=1}^n$ be a basis of L, then there exists $\{v_j\}_{j=1}^n \subset L$, called the **dual basis** such that $B(u_i, v_j) = \delta_{ij}$. (For a proof, see [[Sutherland], Lecture #5, Corollary 5.14, p. 5])

Definition 1.1.13 (Dual Module). Assume the RKLS −Setup, where K and L are number fields. Let T be an R-submodule of L. The Dual module $D(T)$ of T is defined as,

$$D(T) = D_R(T) := \{v \in L \mid B(v, T) \subset R\}.$$

Definition 1.1.14 (The Discriminant ideal $\mathfrak{d}$, The absolute discriminant Δ, The Different $\mathfrak{D}$). Assume the RKLS − Setup.

1. The discriminant ideal $\mathfrak{d}$ is a fractional ideal of R defined as

$$\mathfrak{d} = \mathfrak{d}(S/R) = [D_R(S) : S]_R \in \mathcal{I}_K$$

2. Choose a basis $x_1, \ldots, x_n$ of L over K. The absolute discriminant Δ is defined as

$$\Delta = \Delta(x_1, \ldots, x_n) := \det([T_{L/K}(x_i \cdot x_j)_{ij}]_{1 \le i,j \le n}$$

3. The different $\mathfrak{D} = \mathfrak{D}(S/R)$ is an integral ideal of S defined as the inverse of the dual module of S over R. i.e.,

$$\mathfrak{D} := D_R(S)^{-1} \subset S.$$

An important result is that $\mathfrak{d} = N_{L/K}(\mathfrak{D})$. For a proof, we refer to [[CasFrö], Chapter 1, Corollary 3 of Proposition 4, (p. 16)].

We now discuss some important properties and give some explicit description of the different and the discriminant under the RKLS −Setup. For the proofs, we refer to Chapter 1, Section 4 of [CasFrö].

Assume the RKLS −Setup. Let $x \in S$ so that $L = K[x]$ (existence of such an x follows from the Primitive element theorem). Let $g(X)$ be the minimal polynomial of x over K. The ring $R[x]$ then spans L, and is free as an R-module on $1, x, \ldots, x^{n-1}$. Let $g'(x)$ denote the derivative of $g(X)$. then,

- $\mathfrak{d}(R[x]) = (N_{L/K}(g'(x)) \cdot R$ and $R[x] = S \iff \mathfrak{D}(S/R) = (g'(x)) \cdot S$
- If $L \supset F \supset K$, and if T is the integral closure of R in F, and $m = [L : F]$, then, $\mathfrak{d}(S/R) = \mathfrak{d}(T/R)^m \cdot (N_{F/K}(\mathfrak{d}(S/T)))$.

1.1.4 Ramification

We discuss the concept of ramification. Ramification theory shall play an important role in the proof of the *Kronecker-Weber theorem* (Theorem 2.0.1).

Let R_1 and R_2 be Dedekind Domains with fields of fractions, K_1 and K_2 respectively. Let $\mathfrak{p}_2$ be a non-zero prime ideal of R_2 and suppose that the prime ideal $\mathfrak{p}_1 := \mathfrak{p}_2 \cap R_1$ is also non-zero. We now make a few definitions.

Definition 1.1.15.

1. (Residue Class Field): A residue class field k is the field $k = R/\mathfrak{p}$, where $\mathfrak{p}$ is a non-zero prime ideal of the Dedekind Domain R.
2. (The Residue class degree): The residue class degree $f = f(\mathfrak{p}_2/\mathfrak{p}_1)$ is the degree of the field extension, $[k_2 : k_1]$, where $k_1 := R_1/\mathfrak{p}_1$ is embedded naturally in the field $k_2 := R_2/\mathfrak{p}_2$. i.e.,
$$f(\mathfrak{p}_2/\mathfrak{p}_1) := [k_2 : k_1].$$
3. (Ramification index): Let $v_{\mathfrak{p}_i}$ denote the discrete valuation induced by the non-zero prime ideals $\mathfrak{p}_i$, for $i = 1, 2$. The ramification index $e(\mathfrak{p}_2/\mathfrak{p}_1)$ is defined as
$$e(\mathfrak{p}_2/\mathfrak{p}_1) := v_{\mathfrak{p}_2}(\mathfrak{p}_1 R_2).$$
4. (Unramified extensions): Assume the RKLS −Setup. L is said to be unramified over K if $e(L/K) = 1$ and k_L is separable over k.

   Remark 1.1.16. We note that L is unramified over K if and only if $\mathfrak{d}(L/K) = R$ (See Chapter 1, Section 5, Theorem 1 (p. 21) of [CasFrö]).
5. (Unramified prime over K): A non-zero prime ideal $\mathfrak{P}$ of S is said to be unramified over K if:
   (a) The ramification index of $\mathfrak{P}$ over $\mathfrak{P} \cap R$ has value 1, and
   (b) $S/\mathfrak{P}$ is separable over $R/(\mathfrak{P} \cap R)$
6. (Unramified in L): A non-zero prime ideal $\mathfrak{p}$ of R is said to be unramified in L if all the primes $\mathfrak{P}$ in S which lie above $\mathfrak{p}$ are unramified over K.

We note an important result: Assume the RKLS-Setup. Let $\mathfrak{p}$ be a non-zero prime ideal of R. $\mathfrak{p}$ is unramified in L if and only if $\mathfrak{p}$ does not divide the discriminant $\mathfrak{d}$ of S/R. Thus, *almost all* (i.e., all but finitely many) non-zero prime ideals of R are unramified in S. In fact, the ones which are unramified are precisely all those which do not divide the discriminant.

We shall state few more results which shall be useful later.

Lemma 1.1.17 (Hensel's lemma). Let k be a field which is complete with respect to the Non-Archimedean valuation $|\cdot|$ and let $f(X) \in \mathfrak{o}[X]$, where $\mathfrak{o} := \{x \in k \mid |x| \leq 1\}$ is the ring of integers of the field k. If there exists $a_0 \in \mathfrak{o}$ such that $|f(a_0)| < |f'(a_0)|^2$, then, there is a unique solution $\alpha \in \mathfrak{o}$ to the equation $f(X) = 0$, such that $|\alpha - a_0| = \frac{|f(a_0)|}{|f'(a_0)|} < |f'(a_0)|$ and $|f'(\alpha)| = |f'(a_0)| \neq 0$.

Proof. See [[CasFrö], Appendix C to Chapter 2, (p. 83)]. □

Corollary 1.1.18. Let $g(X)$ be a monic polynomial in $R[X]$ such that $\bar{g}(X)$ is separable. Suppose that $\alpha \in k$ is a root of $\bar{g}(X)$. Then, there is a unique element $x \in R$ such that $g(x) = 0$, and $\bar{x} = \alpha$.

Theorem 1.1.19 (Ostrowski). Let K be a number field. The only non-trivial valuations on K are those equivalent to $|\cdot|_{\mathfrak{p}}$ or the absolute value $|\cdot| = |\cdot|_{\infty}$ which comes from embedding K in $\mathbb{R}$ or $\mathbb{C}$.

Proof. Theorem 3.3 (p. 4) of [Conrad-Ostrowski]. □

Chapter 2

The Kronecker-Weber Theorem

Let $n \in \mathbb{N}$. Let $\zeta_n := e^{\frac{2\pi i}{n}} = \cos(\frac{2\pi}{n}) + i \sin(\frac{2\pi}{n}) \in \mathbb{C}$. The field $\mathbb{Q}(\zeta_n)$ is called the nth *Cyclotomic extension field*.

The Kronecker-Weber theorem explicitly describes abelian extensions of $\mathbb{Q}$ by stating that any finite abelian extension of $\mathbb{Q}$ is contained in some cyclotomic extension. An analogous description for imaginary quadratic number fields is known via the complex multiplication theory of elliptic curves (see [Ghate00] for more details). Hilbert's 12th problem (a.k.a. *Kronecker's Jugendtraum*) presented at the International Congress of Mathematicians at Paris in 1900 asks if a generalization of the Kronecker-Weber theorem is possible using the analogues of the roots of unity for any base number field. In other words, one interpretation of Hilbert's twelfth problem asks to provide a suitable analogue of complex-analytic functions, whose special values would generate the maximal abelian extension K^{ab} of a general number field K.

Though the general case of Hilbert's 12th is still open as of 2022, there has been some progress in this direction (although, $p-$adic analytically,) with a pre-print [D-K21] submitted very recently in March 2021 obtaining an unconditional solution to Hilbert's 12th problem for totally real fields involving $p-$adic integration, for infinitely many primes p. We refer the reader to a master's thesis [Sheth21], which gives give a precise statement of the *Brumer-Stark conjecture* and a detailed exposition of Dasgupta and Kakde's proof of the same (away from the prime $p = 2$) in [D-K20]. The work of Dasgupta and Kakde on the Brumer-Stark conjecture in [D-K20] led them to give an explicit class field theory for totally real number fields in their pre-print [D-K21]. For an exposition and historical details on Hilbert's 12th problem, we refer to Schappacher's article [Schappacher].

The Kronecker-Weber theorem offers us 'Objects' to look out for and aids us in explicitly describing finite abelian extensions of $\mathbb{Q}$. The goal of this chapter is to outline a proof of

the same without using heavy machinery. The main result which we use in our proof is Minkowski's theorem, which, in class-field theoretic terms (see Section 4.4.5 of Chapter 4), says that the Hilbert class field of $\mathbb{Q}$ is $\mathbb{Q}$ itself.

We restate the main theorem of this chapter is as follows:

THEOREM 2.0.1 (Kronecker-Weber). Every finite abelian extension K of $\mathbb{Q}$ is contained in a cyclotomic extension $\mathbb{Q}(\zeta_n)$, for some $n \in \mathbb{N}$.

REMARK 2.0.2. We can in fact compute the positive integer n in Theorem 2.0.1, and it only depends on the prime factorization of the order of the Galois group $\mathrm{Gal}(K/\mathbb{Q})$, as we shall see later. (See Proposition 2.0.10 and Proposition 2.0.11)

For example, to find an abelian degree 3 extension K of $\mathbb{Q}$, we are first led to consider the cyclotomic extension $\mathbb{Q}(\zeta_7)/\mathbb{Q}$. This is because, from Galois theory, the degree of the extension $[K : \mathbb{Q}]$ must be divisible by 3. Now, as the converse of Lagrange's theorem from group theory holds for finite abelian groups (see [MSE-1]), we are assured of an index 3 normal subgroup in $\mathrm{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q})$, and we look for its corresponding fixed field in $\mathbb{Q}(\zeta_7)$, which would be our required degree 3 abelian extension of $\mathbb{Q}$, as a result of the fundamental theorem of Galois theory. Moreover, using Theorem 2.0.4, we get infinitely many degree 3 abelian extensions of $\mathbb{Q}$ as those degree 3 sub-extensions of the cyclotomic extensions $\mathbb{Q}(\zeta_p)$, for primes $p \equiv 1 \mod (3)$.

NOTE 2.0.3. The above arguments we made for the prime number 3 works in fact, for any prime number q, and we similarly get infinitely many degree q sub-extensions of the cyclotomic extensions $\mathbb{Q}(\zeta_p)$, for primes $p \equiv 1 \mod q$. We later extend these ideas to prove the inverse Galois problem for finite abelian groups (see Section 2.0.3).

THEOREM 2.0.4 (Dirichlet's theorem about primes in arithmetic progressions). Let $r \in \mathbb{N}$ and $s \in \mathbb{Z}$ such that $(r, s) = 1$. Then, there are infinitely many primes in the arithmetic progression $tm + s$, for $t \in \mathbb{Z}$.

Proof. See [[Janusz], Chapter IV, Section 5, Theorem 5.9, p. 166], where *Dirichlet's theorem* is proved using the *Frobenius Density Theorem*. □

REMARK 2.0.5. For a proof of the fact that, given a natural number $n > 1$, there are infinitely many primes p of the form $p \equiv 1 \mod n$ (the special case of Theorem 2.0.4 - which is all we actually need), we mention another reference [[Washington], Chapter 2, Corollary 2.11, (p. 13)]. We note that the proof given here is an elementary proof using *cyclotomic polynomials*, and this gives us a special case of Theorem 2.0.4.

We now outline an elementary proof of Theorem 2.0.1. Our main references for this section are the exercises in [Marcus] and the expository articles [Greenberg74], [Branchereau] and [Verser]. Of the many articles on the elementary proof of the Kronecker-Weber theorem, we found the one by Igor Rapinchuk [IR] to be particularly helpful.

2.0.1 Notation and preliminary results

Let K be a number field. Let O_K denote the ring of algebraic integers in K. Let L/K denote a degree n extension of number fields. Given a prime $\mathfrak{p} \in O_K$, consider a prime $\mathfrak{P} \in O_L$ lying over it. Assume further that L/K is Galois with Galois group G.

We note the following:

- $\mathfrak{P}$ and $\sigma(\mathfrak{P})$ lie over $\mathfrak{p}$, and the exponent with which they appear in the prime factorization of the ideal $\mathfrak{p}O_L$ is the same for all $\sigma \in G$. $\mathfrak{P}$ and $\sigma(\mathfrak{P})$ are said to be *conjugate prime ideals* of O_L.
- If $\mathfrak{p}$ is a prime of O_K, then all the primes $\mathfrak{P} \subset O_L$ which lie over $\mathfrak{p}$ are conjugate of each other. They all have the same residual degree f and the same ramification index e. Thus,
$$\mathfrak{p}O_L = \left(\prod_{i=1}^{g} \mathfrak{P}_i \right)^e ,$$
where, g is the total number of distinct conjugates of $\mathfrak{P}$ and also, $n = efg$. (See [[Samuel], Chapter 6, Section 2, Proposition 1, (p. 89)] for a proof.)

2.0.2 Reduction to a Special Case

We now reduce the proof of Theorem 2.0.1 to a Special Case, which is described in Theorem 2.0.6.

Let $K/\mathbb{Q}$ be a finite abelian extension whose Galois group is G. Using the structure theorem of finitely generated abelian groups , write $G = C_1 \times C_2 \times \cdots \times C_l$, where each C_i is a cyclic group of order $p_i^{m_i}$, for some prime number p_i and some positive integer $m_i > 0$, for all $1 \leq i \leq l$.

Set $H_i := C_1 \times C_2 \times \cdots \times C_{i-1} \times C_{i+1} \times \cdots \times C_l$, for all $1 \leq i \leq l$.

Consider the corresponding fixed field $K_i := K^{H_i}$, for all $1 \leq i \leq l$. Then, $K_i/\mathbb{Q}$ is a cyclic extension of degree $p_i^{m_i}$ and $K = K_1 \cdots K_l$. So, if each of $K_i \subset \mathbb{Q}(\zeta_{n_i})$ for some positive integer n_i, then $K \subset \mathbb{Q}(\zeta_n)$, where $n = n_1 \cdots n_l$.

Thus, the proof of the Kronecker Weber theorem for finite abelian extensions of $\mathbb{Q}$

boils down to the case of a cyclic extension $K/\mathbb{Q}$, where K is a cyclic extension of degree p^r for some prime number p and some integer $r \geq 1$. We shall reduce the proof of the Kronecker-Weber theorem to a Special Case which we shall describe below.

Theorem 2.0.6 (Special Case). It is enough to prove the Kronecker-Weber theorem for the case when $K/\mathbb{Q}$ is a finite cyclic extension of degree p^r, for a prime number p, and the only ramified rational prime in K is p.

The proof of Theorem 2.0.6 follows from the following proposition:

Proposition 2.0.7. Consider an abelian extension $K/\mathbb{Q}$ of degree p^r, where p is a prime number. Assume there exists a prime $q \neq p$ which is ramified in K. Then, there exists a sub-field $L \subset \mathbb{Q}(\zeta_q)$ and an abelian extension $K'/\mathbb{Q}$ of degree p^l (for some integer $l \in \mathbb{N}$) such that:

1. $KL = K'L$
2. The unramified primes of K' contain the unramified primes of K.
3. q is unramified in K'.

Proof. See [Section 4, Proposition 8, (p. 13) of [IR]]. □

We note that the proof of Theorem 2.0.1 which we outline in this report mainly uses a fundamental result from the geometry of numbers – the *Minkowski's theorem*, which states the following:

Theorem 2.0.8 (Minkowski's theorem). For any number field $K \neq \mathbb{Q}$, as the absolute discriminant $\Delta(K/\mathbb{Q}) \neq \pm 1$, there is at least one prime $p \in \mathbb{N}$ that ramifies in K.

Proof. See [[Janusz], Chapter I, Section 13, Theorem 13.9, (p. 73)]. □

Now, by induction on the number of rational ramified primes $q \neq p$ in a finite cyclic extension $K/\mathbb{Q}$ of degree p^r, we can eliminate all those finite ramified primes $q \neq p$ using the above proposition (Proposition 2.0.7) and the *Minkowski Theorem*. We are thus ultimately led to prove the Kronecker-Weber theorem only for the Special Case.

We prove the Special Case in two cases: The first when the rational prime p is even and the second when it is odd.

Case I: $p = 2$

Lemma 2.0.9. Let $E/\mathbb{Q}$ be a quadratic extension in which 2 is the only ramified prime. Then, E is one of the fields $\mathbb{Q}(\sqrt{-1}), \mathbb{Q}(\sqrt{-2})$ or $\mathbb{Q}(\sqrt{2})$, all of which are contained in the cyclotomic extension $\mathbb{Q}(\zeta_8)$.

Proof. Let $E = \mathbb{Q}(\sqrt{D})$, for some square-free integer D.
The discriminant $\mathfrak{d}(E) = \pm 2^l$, for some integer $l \in \mathbb{Z}$ as 2 is the only ramified prime. The discriminant $\mathfrak{d}(K)$ of a number field K is defined as $\mathfrak{d}(K) = \det([\omega_i^{(j)}]_{ij})^2$, where $\omega_1, \ldots, \omega_n$ is an integral basis for K and $\omega_i^{(j)}$ denotes the jth-conjugate (other roots of the minimal polynomial of ω over $\mathbb{Q}$) of ω over $\mathbb{Q}$. Note also that $\mathfrak{d}(K) = \det([T_{K/\mathbb{Q}}(x_i \cdot x_j)])$, where $x_1, \ldots, x_n$ is a $\mathbb{Q}$-basis for K. We now explicitly calculate the discriminant. When $D \equiv 2, 3 \mod 4$, $\{1, \sqrt{D}\}$ forms an integral basis for O_K and hence, $\mathfrak{d}(E) = 4D$. When $D \equiv 1 \mod 4$, $\{1, \frac{1+\sqrt{D}}{2}\}$ forms an integral basis for O_K and hence, $\mathfrak{d}(E) = D$. Therefore, $D \in \{-1, \pm 2\}$. The quadratic extensions $\mathbb{Q}(\sqrt{-1}), \mathbb{Q}(\sqrt{-2}), \mathbb{Q}(\sqrt{2})$ are all contained in the cyclotomic extension $\mathbb{Q}(\zeta_8)$ as $\zeta_8^2 = e^{\frac{2\pi i}{8} \cdot 2} = \sqrt{-1}$ and $(\zeta_8 \pm \zeta_8)^2 = \pm 2$. □

Proposition 2.0.10. Let $K/\mathbb{Q}$ be an abelian extension of degree 2^m, where m is a positive integer. Assume further that the only ramified rational prime in K is 2. Then, $K \subset \mathbb{Q}(\zeta_{2^{m+2}})$.

Proof. We have proved the case $m = 1$ in Lemma 2.0.9. Now, we assume that $m > 1$.

Let $M := \mathbb{Q}(\zeta_{2^{m+2}})$ and $L := M^\sigma$, where σ is the complex conjugation automorphism.

$\Rightarrow L/\mathbb{Q}$ is cyclic of degree 2^m. As L is totally real, as a consequence of Lemma 2.0.9, $\sqrt{2} \in L$. Let $\langle \tau \rangle = \mathrm{Gal}(L/\mathbb{Q})$. Let $\tilde{\tau}$ be any extension of τ to an element of $\mathrm{Gal}(KL/\mathbb{Q})$. i.e., $\tilde{\tau}_{|_L} = \tau$. $[K : \mathbb{Q}] = [L : \mathbb{Q}] = 2^m$

From the embedding $\mathrm{Gal}(KL/\mathbb{Q}) \to \mathrm{Gal}(K/\mathbb{Q}) \times \mathrm{Gal}(L/\mathbb{Q})$, we get that the order of any element $\sigma \in \mathrm{Gal}(KL/\mathbb{Q})$ divides 2^m. Now, as $\tilde{\tau}_{|_L} = \tau$ is of order 2^m, the order of τ is exactly 2^m . Define $F := (KL)^{\tilde{\tau}}$. Then, $[KL : F] = 2^m$. As τ generates $\mathrm{Gal}(L/\mathbb{Q})$, $L^\tau = \mathbb{Q}$.

$\therefore F \cap L = \mathbb{Q}$. Now consider $KL/\mathbb{Q}$. $[KL : \mathbb{Q}]$ is a power of 2. $[F : \mathbb{Q}]$ is a power of 2. We now use the following result:

For any prime number p and any integer $r \geq 1$, the only ramified rational prime in $\mathbb{Q}(\zeta_{p^r})$ is p which is totally ramified as well. (We compute the discriminant of $\mathbb{Q}(\zeta_{p^r})$ and see that its only prime divisor is p)

Now, 2 is the only ramified prime in K and L. Moreover, 2 is the only ramified prime in F as well. Define $F_0 := F^\sigma$. Claim: $F_0 = \mathbb{Q}$. Assume the contrary that $F_0 \neq \mathbb{Q}$. Then, using previous lemma, $\sqrt{2} \in F_0 \Rightarrow \sqrt{2} \in L \cap F = \mathbb{Q} \quad \to\leftarrow \therefore F_0 = \mathbb{Q}$. Thus, $[F : \mathbb{Q}] = 1$ or 2. Note that $[KL : F][F : \mathbb{Q}] = [KL : \mathbb{Q}]$ If $[F : \mathbb{Q}] = 2$, then $F = \mathbb{Q}(\sqrt{-1})$ or $\mathbb{Q}(\sqrt{-2})$, which are both contained in M. In this case, $[KL : \mathbb{Q}] = 2^{m+1}$ and as F is complex, F is not contained in the totally real field L, and hence, $K \subset KL = FL = M$. If $[F : \mathbb{Q}] = 1$, then $[KL : \mathbb{Q}] = 2^m = [L : \mathbb{Q}] \Rightarrow K = L \subset M$. □

Case II: p is odd

Proposition 2.0.11. Let p be an odd prime. Let $K/\mathbb{Q}$ be an abelian extension of degree p^m such that the only ramified rational prime is p. Then, $K \subset \mathbb{Q}(\zeta_{p^{m+1}})$.

Proof. The proof is parallel to the previous case as in Proposition 2.0.10. For details, see [IR].

$M := \mathbb{Q}(\zeta_{p^{m+1}})$. Let $L \subset \mathbb{Q}(\zeta_{p^{m+1}})$ be the unique sub-extension of degree p^m over $\mathbb{Q}$. Then, $L/\mathbb{Q}$ is cyclic and only ramified at the prime p. Let $\langle\tau\rangle = \mathrm{Gal}(L/\mathbb{Q})$. Let $\tilde{\tau}$ be any extension of τ to an element of $\mathrm{Gal}(KL/\mathbb{Q})$. i.e., $\tilde{\tau}_{|_L} = \tau$. From the natural embedding $\mathrm{Gal}(KL/\mathbb{Q}) \to \mathrm{Gal}(K/\mathbb{Q}) \times \mathrm{Gal}(L/\mathbb{Q})$, we conclude that the order of $\tilde{\tau}$ is exactly p^m. Define $F := (KL)^{\tilde{\tau}}$. Then, $[KL : F] = p^m$. As τ generates $\mathrm{Gal}(L/\mathbb{Q})$, $\quad L^{\tau} = \mathbb{Q}$. $\therefore F \cap L = \mathbb{Q}$. Now consider $KL/\mathbb{Q}$. $\quad [KL : \mathbb{Q}]$ is a power of 2.

We, like in Proposition 2.0.10, use a similar result: *There exists a unique cyclic extension $P/\mathbb{Q}$ of degree p such that only p is the ramified rational prime.* (For the proof, see [[IR], Section 6, Corollary 7, p. 16].)

We now claim that: $F = \mathbb{Q}$.

Assume the contrary that $F \neq \mathbb{Q}$. Then, there exists a sub-extension $F_0 \subset F$ of degree p over $\mathbb{Q}$. Only p is ramified in K and L and hence, only p is ramified in KL as well. Therefore, only p is ramified in $F_0 \subset F = (KL)^{\tilde{\tau}}$. Similarly, we show that there exists $L_0 \subset L$ having degree p over $\mathbb{Q}$. $\therefore F_0 = L_0 \subset F \cap L = \mathbb{Q}$. $\quad \to\leftarrow$

Thus, $F = \mathbb{Q}$. Therefore, $[KL : \mathbb{Q}] = p^m = [L : \mathbb{Q}]$. $\Rightarrow K \subset L$. □

2.0.3 An application: IGP for finite abelian groups

The inverse Galois problem (IGP) asks whether or not every finite group appears as the Galois group of some Galois extension of $\mathbb{Q}$. We can prove the following theorem as a consequence of the Kronecker-Weber theorem. We can solve the IGP for finite abelian groups without using the Kronecker-Weber theorem (for example using *Dirichlet's theorem about primes in arithmetic progressions for primes $p \equiv 1 \mod n$* (see [Tiesinga], where detailed proofs are given). We however, remark that using the Kronecker-Weber theorem in addition, we get the following result:

"Given any finite abelian group G, we explicitly find a cyclotomic extension $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ which contains a sub-extension whose Galois group is isomorphic to G. Moreover, the positive integer n which we obtain only depends on the order of G."

We wish to prove the following:

THEOREM 2.0.12. (*IGP for finite abelian groups*) *Any finite abelian group is the Galois group of some Galois extension of $\mathbb{Q}$.*

We shall first prove the IGP for finite cyclic extensions of $\mathbb{Q}$.

As a result of the *Kronecker-Weber theorem* proved in the previous section, we have the

following:

1. Let p be an odd prime. Let $K/\mathbb{Q}$ be an abelian extension of degree p^m. Then, $K \subset \mathbb{Q}(\zeta_{p^{m+1}})$. (Proposition 2.0.11)
2. Let $K/\mathbb{Q}$ be an abelian extension of degree 2^m, where m is a positive integer. Then, $K \subset \mathbb{Q}(\zeta_{2^{m+2}})$ (Proposition 2.0.10)

Reduction of Theorem 2.0.12 to finite cyclic extensions

Case I: $p = 2$

Aim: Find a cyclic extension $K/\mathbb{Q}$ of order 2^m (for given positive integer $m > 0$).

Consider $L_p := \mathbb{Q}(\zeta_{2^{m+2}})$. For any $r \geq 2$, the Galois group of the cyclotomic extension $\mathbb{Q}(\zeta_{2^r})/\mathbb{Q}$ is isomorphic to $(\mathbb{Z}/(2^r)\mathbb{Z})^\times \cong \mathbb{Z}/(2^r - 2^{r-1})\mathbb{Z}$.

Moreover, $(\mathbb{Z}/(2^r)\mathbb{Z})^\times \cong \{\pm 1\} \times \langle \bar{5} \rangle$ is of order 2^{r-1}, where $\langle \bar{5} \rangle \leqslant (\mathbb{Z}/(2^r)\mathbb{Z})^\times$ is the subgroup of order 2^{r-2} generated by the element $5 \mod (2^r)$ Thus, the subgroup $\langle \bar{5} \rangle \leqslant (\mathbb{Z}/(2^{m+2})\mathbb{Z})^\times$ is of order m and the subgroup $\{\pm 1\} \leqslant (\mathbb{Z}/(2^{m+2})\mathbb{Z})^\times$ is of index m. Thus, we now consider the fixed field $K := \mathbb{Q}(\zeta_{2^{m+2}})^{\langle \sigma \rangle}$ of the complex conjugation σ in $\mathbb{Q}(\zeta_{2^{m+2}})$. (Note that K is totally real). Then, $K/\mathbb{Q}$ is the required cyclic extension of $\mathbb{Q}$ of order 2^m. △

Case II: p is an odd prime

Aim: Find a cyclic extension $K/\mathbb{Q}$ of order p^m (for given positive integer $m > 0$).

Consider $L_p := \mathbb{Q}(\zeta_{p^{m+1}})$. For an odd prime p and any positive integer $r \geq 1$, $\mathrm{Gal}(\mathbb{Q}(\zeta_{p^r})) \cong (\mathbb{Z}/p^r\mathbb{Z})^\times$ is cyclic of order $p^r - p^{r-1}$. Let p be an odd prime. Let $K/\mathbb{Q}$ be an abelian extension of degree p^m. Then, $K \subset \mathbb{Q}(\zeta_{p^{m+1}})$. Thus, as $\mathrm{Gal}(\mathbb{Q}(\zeta_{p^{m+1}})/\mathbb{Q})$ is cyclic of order $p^m(p-1)$, we find an index p^m subgroup $H \leqslant G$ and consider the fixed field $K := \mathbb{Q}(\zeta_{p^{m+1}})^H$. Then, $K/\mathbb{Q}$ is the required cyclic extension of $\mathbb{Q}$ of order p^m. △

Proof of Theorem 2.0.12

Statement: Any finite abelian group is the Galois group of some Galois extension of $\mathbb{Q}$. (We in fact, find a Galois extension which is contained in a cyclotomic extension $\mathbb{Q}(\zeta_n)$, where n depends on the order of the given finite abelian group.)

Proof. Let G be a finite abelian group.

Say $G = C_1 \times C_2 \times \cdots \times C_l$, where C_i is a cyclic group of order $p_i^{m_i}$, for prime numbers p_i and positive integers m_i, $\forall\, 1 \leq i \leq l$. If p_i is even, let $L_{p_i} := \mathbb{Q}(\zeta_{2^{m_i+2}})$ and if p is

odd, let $L_{p_i} := \mathbb{Q}(\zeta_{p^{m_i+1}})$, $\forall\, 1 \leq i \leq l$. Also, let $G_i = \mathrm{Gal}(L_{p_i}/\mathbb{Q})$, $\forall\, 1 \leq i \leq l$. As we have shown before, we find subgroups $H_i \leqslant G_i$ such that $\mathrm{Gal}(L_{p_i}^{H_i}/\mathbb{Q}) \cong C_i$, $\forall\, 1 \leq i \leq l$. Define $K_i := L_{p_i}^{H_i}$. Consider $K := K_1 K_2 \cdots K_l$ If p and q are distinct rational primes, then $\mathbb{Q}(\zeta_{p^r}) \cap \mathbb{Q}(\zeta_{q^s}) = \mathbb{Q}$, for all positive integers r and s. Thus, using Galois theory, $\mathrm{Gal}(K/\mathbb{Q}) \cong \mathrm{Gal}(K_1/\mathbb{Q}) \times \cdots \times \mathrm{Gal}(K_l/\mathbb{Q}) \cong C_1 \times C_2 \times \cdots \times C_l \cong G$. □

Chapter 3

Number fields

Number fields, as discussed before, are nothing but finite abelian extensions of the field of rational numbers $\mathbb{Q}$. In this chapter, we discuss some of their properties. The two main results in this chapter are about the structure of the number rings which is given by Dirichlet's unit theorem, and the other is about the Minkowski's bound – which is a tool from the geometry of numbers which was used extensively in the proof of Kronecker-Weber theorem in Chapter 2. We also use the latter and discuss some methods to compute ideal class groups of some small degree number fields.

3.0.1 Dirichlet's Unit Theorem

Let K be a number field and $\mathcal{O}_K$ number ring. The set of all units in $\mathcal{O}_K$ forms a multiplicative subgroup of $K^\times$. We state an almost complete description of the structure of the unit group of K, known as *Dirichlet's Unit theorem*. We refer to [Murthy-Esmonde] and [Janusz] for proofs and details. Theorem 3.0.1, which we state below, is very useful in computing the ideal class groups of some number fields and finding solutions to some diophantine equations.

Theorem 3.0.1 (Dirichlet's Unit theorem). Let U_K be the unit group of K. Let $n := [K : \mathbb{Q}]$ and we write $n = r_1 + 2r_2$, where, r_1 and $2r_2$ are the number of real and non–real embeddings of K in $\mathbb{C}$ respectively. Then, there exist units $\epsilon_1, \ldots, \epsilon_r$ in U_K, where $r = r_1 + r_2 - 1$, such that every unit $\epsilon \in U_K$ can be expressed uniquely as $\epsilon = \zeta \epsilon_1^{n_1} \cdots \epsilon_r^{n_r}$, for some integers $n_1, \ldots, n_r$ and a root of unity ζ in $\mathcal{O}_K$.

In other words, if W_K is the subgroup of $\mathcal{O}_K$ consisting of roots of unity occurring in $\mathcal{O}_K$, then W_K is a finite cyclic group[1], and $U_K \simeq W_K \times \mathbb{Z}^r$.

[1] W_K is in fact, also of even order due to a result of Kronecker. See [[Murthy-Esmonde], Chapter 1, Section 8.1, Exercise 8.1.2].

Proof. See [[Janusz], Chapter 1, Section 13, Theorem 13.12, (p. 77)]. □

3.0.2 General results about number fields

Proposition 3.0.2. Let $\mathbb{Q}(\zeta_n)$ denote the nth Cyclotomic extension for some $n \in \mathbb{N}$. We then have,

1. The ring of integers of $\mathbb{Q}(\zeta_n)$ is $\mathbb{Z}[\zeta_n]$. (See [Washington], Chapter 2, Theorem 2.6, p. 11)

2. $d_{\mathbb{Q}(\zeta_n)/\mathbb{Q}} = (-1)^{\frac{\phi(n)}{2}} \cdot \frac{n^{\phi(n)}}{\prod_{p|n} p^{\phi(n)/(p-1)}}$. (See [Washington], Chapter 2, Proposition 2.7, p. 12)

3. The ring of integers of the maximal real sub-field $\mathbb{Q}(\zeta_n)^+ := \mathbb{Q}(\zeta_n + \zeta_n^{-1})$ of $\mathbb{Q}(\zeta_n)$ is $\mathbb{Z}[\zeta_n + \zeta_n^{-1}]$. (See [Washington], Chapter 2, Proposition 2.16, p. 16)

Proposition 3.0.3.

$$\begin{array}{c} M \\ | \\ L \\ | \\ K \end{array}$$

Let $K \subset L \subset M$ be a tower of number fields. Then,

$$d_{M/K} = d_{L/K}^{[M:L]} \cdot N_{L/K}(d_{M/L})$$

where, $d_{M/K}$ denotes the discriminant of the extension M/K .

Proof. See [[Neukirch] Chapter III, Corollary 2.10. (p.)]. □

Proposition 3.0.4. Assume the RKLS Setup. If $S = R[\alpha]$ for some $\alpha \in L$, and if $f(X) \in R[X]$ is the minimal polynomial of α over K , then, the different

$$\mathfrak{D}_{L/K} = (f'(\alpha))$$

Hence, the discriminant ideal $\mathfrak{d}_{L/K} = N_{L/K}(\mathfrak{D}_{L/K}) = N_{L/K}((f'(\alpha)))$

Proof. See □

Cyclotomic Extensions of the type $\mathbb{Q}(\zeta_p)$, for a prime number p.

Proposition 3.0.5. Let p denote a prime number. Let $\zeta_p := e^{\frac{2\pi i}{p}} \in \mathbb{C}$ denote the primitive p^{th}−root of unity. Consider the cyclotomic extension $K := \mathbb{Q}(\zeta_p)/\mathbb{Q}$. Let $G := \mathrm{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q}) \cong (\mathbb{Z}/p\mathbb{Z})^\times$ denote its Galois group. Then,

1. The different ideal $\mathfrak{D}$ of $\mathbb{Q}(\zeta_p)$ is given by $\mathfrak{D} = (\frac{p}{\zeta_p - 1})$
2. The only ramified rational prime $q \in \mathbb{Z}$ in K is $q = p$. (In fact, p is totally ramified.)

Proof. See [[IR], Section 3, Proposition 7, (p. 12)]. □

3.1 Minkowski's bound and the ideal class group

Let K be a number field. We know that any ideal class in the class group (or the ideal class group) of K contains an integral ideal with norm bounded above by the Minkowski bound $M = \frac{n!}{n^n} \cdot \left(\frac{4}{\pi}\right)^{r_2} \cdot \sqrt{|d_K|}$, where $n = [K : \mathbb{Q}]$, r_2 is the number of pairs of distinct non-real embeddings of K in $\mathbb{C}$, and d_K is the discriminant of K over $\mathbb{Q}$. (See [[Murthy-Esmonde], Chapter 6, Section 6.5, Exercise 6.5.12, (p. 78)])

For any non-zero (fractional) ideal $\mathfrak{a} \in O_K$, we denote by $[\mathfrak{a}]$ its ideal class in the class group $\mathrm{Cl}(K) := \mathscr{I}_K/\mathscr{P}_K$, where $\mathscr{I}_K$ is the group of non-zero fractional ideals of O_K and $\mathscr{P}_K$ is the subgroup of $\mathscr{I}_K$ generated by the non-zero principal fractional ideals.

$\sim$ is the equivalence relation that defines the class group. i.e,

$$\mathrm{Cl}(K) = \mathscr{I}_K / \sim$$

For any two fractional ideals $\mathfrak{a}, \mathfrak{b} \in \mathscr{I}_K$, we use the notation $\mathfrak{a} \sim \mathfrak{b}$ to mean that $\mathfrak{b} = \gamma\mathfrak{a}$, for some $\gamma \in K^\times$. We shall use the symbol $\mathbb{1}$ to denote the identity of the ideal class group.

Remark 3.1.1. The notion of an ideal class group $\mathrm{Cl}(R)$ can similarly be defined for any Dedekind domain R as the quotient group $\mathscr{I}_R/\mathscr{P}_R$ of the fractional ideal group $\mathscr{I}_R$ of R by the subgroup $\mathscr{P}_R$ of non-zero principal fractional ideals of R.

Due to a theorem of Claborn, it is known that any abelian group can be realized as the (ideal) class group of some Dedekind domain (see [Claborn]). But, the analogous problem in this direction is open for rings of algebraic integers of number fields. In other words, it is not known whether every abelian group can be realized as the class group of some number field (see '*Remarks on Dedekind domains and class groups*' (p. 358) in [PSS]).

3.1.1 Method of calculation of class group

We describe an important algorithm below which shall help us in computing ideal class groups of number fields.

Proposition 3.1.2. Let K be a number field. Suppose that α is a primitive for K. i.e., $K = \mathbb{Q}(\alpha)$.

(a) Let $\mathfrak{q}$ be a prime in $\mathcal{O}_K$ which lies over a rational prime $q \in \mathbb{Z}$. If $f(\mathfrak{q}|q) = 1$, then $\exists\, 0 \leq r < q$ such that $q \mid N_{K/\mathbb{Q}}(\alpha + r)$.

(b) If there exists a *unique* prime $\mathfrak{q} \subset \mathcal{O}_K$ which lies over a rational prime q such that $f(\mathfrak{q}|q) = 1$, then for all $r, s \in \mathbb{F}_q$ such that $r \neq s$, $q \nmid N_{K/\mathbb{Q}}(\alpha + r)$ or $q \nmid N_{K/\mathbb{Q}}(\alpha + s)$.

(b) says that if the divisibility $q \mid N_{K/\mathbb{Q}}(\alpha + r)$ in (a) happens for more than once, (i.e., for more than one r), then the corresponding primes which lie over q cannot be the same.

(c) If there exists a unique $0 \leq r < q$ such that $q \mid N_{K/\mathbb{Q}}(\alpha + r)$, then there exists a *unique* prime $\mathfrak{q} \in \mathcal{O}_K$ such that $\mathfrak{q}|q$ and $f(\mathfrak{q}|q) = 1$.

Proof. Suppose there exists a prime $\mathfrak{q}$ in $\mathcal{O}_K$ which lies over a rational prime q such that $f(\mathfrak{q}|q) = 1$. i.e, its residue field is still $\mathbb{F}_q$ and $[\mathcal{O}_K/\mathfrak{q} : \mathbb{Z}/(q)] = 1$. We have that $\mathcal{O}_K/\mathfrak{q} \cong \mathbb{Z}/(q)$, $N_{K/\mathbb{Q}}(\mathfrak{q}) = q$ and $|\mathcal{O}_K/\mathfrak{q}| = q$.

We see that the following is true: $\alpha + r \equiv 0 \mod \mathfrak{q}$, *for some* $0 \leq r < q$.

We see that $(\alpha + r) \mod \mathfrak{q} \in \mathbb{Z}/(q) \Rightarrow \mathfrak{q} \mid (\alpha + r) \Rightarrow q \mid N_{K/\mathbb{Q}}(\alpha + r)$.

On the contrary, if $q \mid N_{K/\mathbb{Q}}(\alpha + r)$ and $q \mid N_{K/\mathbb{Q}}(\alpha + s)$, then $\mathfrak{q}_r \neq \mathfrak{q}_s$, where $\mathfrak{q}_r \mid (\alpha + r)$ and $\mathfrak{q}_s \mid (\alpha + s)$, for primes $\mathfrak{q}_r$ and $\mathfrak{q}_s$ lying over q.

Because, otherwise, if $\mathfrak{q}_r = \mathfrak{q}_s$, then $\mathfrak{q}_i \mid (\alpha + r - \alpha - s) = (r - s)$, for all $i = 1, 2$. We note that $(r - s) \in \mathbb{F}_q^{\times}$. i.e., $r - s$ is a unit modulo q, which is a contradiction. □

Theorem 3.1.3 (Dedekind's criterion).

1. Let K be a number field. Let $\theta \in K$ such that $\mathcal{O}_K = \mathbb{Z}[\theta]$. Let $m_\theta(X) \in \mathbb{Z}[X]$ denote the minimal polynomial of θ over $\mathbb{Q}$. For any prime number p, if $f(X) = f_1(X)^{e_1} \cdots f_l(X)^{e_l} \mod p$ denotes the factorization of $f(X)$ modulo p for irreducible polynomials $f_1(X), \ldots, f_l(X)$, then,

$$p \cdot \mathcal{O}_K = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_l^{e_l},$$

where $\mathfrak{p}_i = ((p, f_i(\theta)))$ are the finite primes of $\mathcal{O}_K$, $\forall\, 1 \leq i \leq l$ and,

$$N_{K/\mathbb{Q}}(\mathfrak{p}_i) = p^{\deg(f_i(X))}$$

2. Let K be a number field. Let $\theta \in K$ and let $m_\theta(X) \in \mathbb{Z}[X]$ denote the minimal polynomial of θ over $\mathbb{Q}$. Let $\Delta(f)$ denote the discriminant of f. i.e., if $f(X) = \prod_{i=1}^{d}(X - \alpha_i) \in \mathbb{C}[X]$, then $\Delta(f) := \prod_{1 \le i < j \le d}(\alpha_i - \alpha_j)$.

   For any prime number p such that $p \nmid \Delta(f)$, if $f(X) = f_1(X)^{e_1} \cdots f_l(X)^{e_l} \mod p$ denotes the factorization of $f(X)$ modulo p for irreducible polynomials $f_1(X), \ldots, f_l(X)$, then,

   $$p \cdot \mathcal{O}_K = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_l^{e_l},$$

   where $\mathfrak{p}_i = ((p, f_i(\theta)))$ are the finite primes of $\mathcal{O}_K$, $\forall\, 1 \le i \le l$ and,

   $$N_{K/\mathbb{Q}}(\mathfrak{p}_i) = p^{\deg(f_i(X))}$$

Proof. See [[Murthy-Esmonde], Chapter I, Section 5.5, Theorem 5.5.1 and Exercise 5.5.2, (p. 65-66)]. □

Chapter 4

Class Field Theory

A *number field* is a finite extension of $\mathbb{Q}$ and a *global function field* is a finite extension of $\mathbb{F}_p(t)$ for some prime number p. A *global field* is either a number field or a global function field. The aim of (abelian) class field theory is to describe the abelian extensions of local or global fields in terms of the arithmetic of the fields themselves.

For an excellent discussion of the history of the class field theory, we refer the reader to K. Conrad's expository article [Conrad-CFT-History]. Though a *non-abelian class field theory* is still not known, Robert Langlands has proposed a web of far-reaching conjectures (known as the *Langlands program*) which among many other things, also contains a non-abelian class field theory. We refer the reader to [Cogdell], [Wyman], [Weinstein], [Lemmermeyer] and [Gelbart] for more details.

Our main reference for this chapter is [Janusz]. The other approaches to class field theory are the algebraic/cohomological approach of Chevalley, Artin and Tate, the explicit approach of Lubin and Tate in the local case and the analytic approach in the global case, all of which are explained in detail in Milne's notes [Milne-CFT].

4.1 Completions

Let K be a number field. We make the following definitions:

Definition 4.1.1 (prime).

1. An equivalence class of valuations on K is called as a *prime* of K.
2. An equivalence class of Archimedean valuations on K is called an *infinite prime* of K.
3. An equivalence class of Non-Archimedean valuations is called as a *finite prime* of K.

As the completion of K at any two equivalent valuations gives rise to isomorphic fields, the term "*completion of K at a prime*" is well-defined and is used to mean the completion of K at any valuation in the prime. Now, as a consequence of Ostrowski's theorem (Theorem 1.1.19), we have that:

Any Archimedean valuation of an algebraic number field K is equivalent to $|\cdot|_1$ defined as $|x|_1 := |\phi(x)|$, where ϕ is a real or complex embedding of K, and $|\cdot|$ is the usual absolute value on $\mathbb{C}$.

Theorem 4.1.2 (Archimedean Valuations of Algebraic Number Fields). Let K be an Algebraic number field. Let $\sigma_1, \ldots, \sigma_r$ denote the distinct embeddings of K into the real number field $\mathbb{R}$, and let $\sigma_{r+1}, \ldots, \sigma_s$ denote the complex embeddings, one from each pair of conjugate embeddings of K into the complex field $\mathbb{C}$. Then, an Archimedean valuation of K is equivalent to **exactly one** of the valuations defined by $|x|_i := |\sigma_i(x)|$, for, $1 \leq i \leq r + s$.

Proof. See [[Janusz], Chapter II, Section 4, Theorem 4.5, p. 111]. □

Definition 4.1.3 (The two types of infinite primes – Real and Complex).

1. An infinite prime of K is called a *real prime* if the completion of K at that prime is the real field $\mathbb{R}$.

2. Similarly, a *complex prime* is an infinite prime of K such that the completion of K at the prime is the complex field $\mathbb{C}$.

Note 4.1.4. Thus, from Theorem 4.1.2, the real primes of an algebraic number field K correspond to the distinct real embeddings of K. And, the complex primes correspond to the conjugate pairs of complex embeddings of K.

We shall follow the following conventions:

Let K be an algebraic number field. Let $\mathfrak{p}$ be an infinite prime of K. Let L/K be a finite extension of Number fields. Let $\mathfrak{P}_1, \ldots, \mathfrak{P}_g$ be the primes of L that extend $\mathfrak{p}$. We define the **ramification index** e_i as follows:

$$e_i := e(\mathfrak{P}_i/p) = \begin{cases} 2 & \text{if } \mathfrak{p} \text{ is real and } \mathfrak{P} \text{ is complex} \\ 1 & \text{if otherwise} \end{cases}$$

We define the *inertia degree* f_i as $f_i = f(\mathfrak{P}_i/\mathfrak{p}) = 1$ in all cases. Note that the following is always true:

$$\sum_{i=1}^{g} e_i f_i = [L : K]$$

DEFINITION 4.1.5 (The Decomposition group D of $\mathfrak{P}$). The Decomposition group $D = D_{\mathfrak{P}}$ of $\mathfrak{P}$ is defined as $D_{\mathfrak{P}} := \{\sigma \in G \mid \sigma(\mathfrak{P}) = \mathfrak{P}\}$.

Let g denote the conjugates of $\mathfrak{P}$. then, $g = |G/D| = \frac{|G|}{|D|}$. Thus, we get $|D| = \frac{n}{g} = ef$.

DEFINITION 4.1.6 (Inertia subgroup I of D). The inertia subgroup $I = I_{\mathfrak{P}} \leqslant D$ of $\mathfrak{P}$ is defined as

$$I := \{\sigma \in D \mid \sigma(x) - x \in \mathfrak{P},\ \forall x \in \mathcal{O}_L.\}$$

Now, we observe the following:

Let L/K be a finite extension of Number fields and let F/K be a Galois extension. Then, the Galois group $G := \mathrm{Gal}(L/K)$ acts on the $F-$algebra $F \otimes_K L$ as,

$$\sigma(\sum a_i \otimes x_i) \to \sum a_i \otimes \sigma(x_i), \forall\, \sigma \in G$$

Moreover, the sub-ring of $G-$fixed points is equal to $F \otimes 1$. (See [[Janusz], Chapter III, Section 1, Lemma 1.1, (p. 121)] for details.)

Thus, in the case when $F = K_{\mathfrak{p}}$, we get the following result:

THEOREM 4.1.7. Let L/K be a Galois extension of number fields. Let $\mathfrak{P}$ be a prime of L, extending a prime $\mathfrak{p}$ of K. Let $D_{\mathfrak{P}}$ denote the decomposition group of $\mathfrak{P}$ in $\mathrm{Gal}(L/K)$. Then, $L_{\mathfrak{P}}$ is Galois over $K_{\mathfrak{p}}$ with the Galois group $D_{\mathfrak{P}}$.

Proof. See [[Janusz], Chapter III, Section 1, Theorem 1.2, (p. 122)]. □

4.2 The Frobenius automorphism

Let $k = \mathbb{F}_q$ be a finite field of characteristic p, where $q = p^f$, for some $f \geq 1$. We define the Frobenius map $\mathrm{Frob}_p : k \to k$ as $\mathrm{Frob}_p(a) = a^p$, for all $a \in k$. Then, $k/\mathbb{F}_p$ is a Galois extension and $\mathrm{Gal}(k/\mathbb{F}_p) = \langle \mathrm{Frob}_p \rangle$.

Suppose that K is a number field such that $K/\mathbb{Q}$ is a Galois extension. Fix a finite prime $\mathfrak{P} \subset \mathcal{O}_K$ lying over a rational prime $p \in \mathbb{Z}$. Let $D_{\mathfrak{P}} = \{\sigma \in G | \sigma(\mathfrak{P}) = \mathfrak{P}\}$ denote the decomposition group of $\mathfrak{P}$. Let $k_{\mathfrak{P}} = \mathcal{O}_K/\mathfrak{P}$. Then, there exists an exact sequence

$$1 \longrightarrow I_{\mathfrak{P}} \longrightarrow D_{\mathfrak{P}} \longrightarrow \mathrm{Gal}(k_{\mathfrak{P}}/\mathbb{F}_p) \tag{4.1}$$

where, $I_{\mathfrak{P}}$ is the inertia group of $\mathfrak{P}$, a subgroup of $D_{\mathfrak{P}}$.

Let L/K be a Galois extension of number fields. Let $\mathfrak{p}$ be a finite prime of $\mathcal{O}_K$ which is unramified in L. If $\mathfrak{P}$ is a finite prime of $\mathcal{O}_L$ lying over $\mathfrak{p}$, then there exists a unique element

$\sigma \in \mathrm{Gal}(L/K)$ such that, for all $\alpha \in \mathcal{O}_L$,

$$\sigma(\alpha) = \alpha^{\mathcal{N}(p)} \mod \mathfrak{P},$$

where $\mathcal{N}(\mathfrak{p}) = |\mathcal{O}_K/\mathfrak{p}|$ is the counting norm.

As $D_{\mathfrak{P}} \simeq \mathrm{Gal}(k_L/k)$, the decomposition group $D_{\mathfrak{P}} = \langle\sigma\rangle$ is a cyclic group generated by the Frobenius automorphism. In the case when $\mathfrak{P}/\mathfrak{p}$ is unramified, we write *the Frobenius automorphism* σ of $\mathfrak{P}$ as $\sigma = \left[\frac{L/K}{\mathfrak{P}}\right]$.

An important property of the Frobenius automorphism is that an unramified prime $\mathfrak{p}$ of a number field K splits completely in the finite extension L/K if and only if $\left[\frac{L/K}{\mathfrak{P}}\right] = 1$.

4.3 The Artin Map

Assume that L/K is an abelian extension of number fields with $G = \mathrm{Gal}(L/K)$. If $\mathfrak{p}$ is an unramified prime of K, then the Frobenius automorphism is same for all the primes $\mathfrak{P}$ of L which lie over $\mathfrak{p}$. Thus, we write $\left[\frac{L/K}{\mathfrak{p}}\right]$ and call this as the *Frobenius automorphism of* $\mathfrak{p}$ to mean $\left[\frac{L/K}{\mathfrak{P}}\right]$, for any prime $\mathfrak{P}$ extending $\mathfrak{p}$.

Let $\mathcal{I}_K$ denote the group of fractional ideals of K. Let $\mathcal{O}_K$ denote the ring of algebraic integers in K. Let S denote the set of all primes in $\mathcal{O}_K$ which are ramified in L. (Note that S is finite). We define a subgroup $\mathcal{I}_K^S$ of $\mathcal{I}_K$, which is generated by all the non-zero primes of $\mathcal{O}_K$ which lie outside S. i.e., a general element $\mathfrak{U}$ of $\mathcal{I}_K^S$ is of the form

$$\mathfrak{U} = \prod_{\substack{\mathfrak{p}\text{ prime} \\ \mathfrak{p}\notin S}} \mathfrak{p}^{a(\mathfrak{p})},$$

where only finitely many exponents $a(\mathfrak{p})$ are non-zero for any fractional ideal $\mathfrak{U}$ of $\mathcal{I}_K$.

We now define the Artin map $\phi_{L/K}$ as follows:

$$\phi_{L/K} : \mathcal{I}_K^S \to \mathrm{Gal}(L/K)$$

$$\phi_{L/K} : \mathfrak{U} \mapsto \prod_{\substack{\mathfrak{p}\text{ prime} \\ \mathfrak{p}\notin S}} \left[\frac{L/K}{\mathfrak{p}}\right]^{a(\mathfrak{p})}$$

Whenever $\mathfrak{p}$ is an unramified prime of K, $\phi_{L/K}(\mathfrak{p})$ is nothing but the Frobenius automorphism $\left[\frac{L/K}{\mathfrak{P}}\right]$.

REMARK 4.3.1. Our main task is to derive the image and kernel of the Artin map in some special cases.

It is relatively easier to compute the kernel and image of the Artin map for cyclotomic fields. We note the same below:

THEOREM 4.3.2. Consider the cyclotomic extension $\mathbb{Q}(\zeta_m)$, for some positive integer m. Let S denote the set of all prime ideals of $\mathbb{Z}$ dividing (m). If $\phi_{L/K} : \mathbb{Q}(\zeta_m) \to \mathrm{Gal}(\mathbb{Q}(\zeta_m))$ is the Artin map, then the map is surjective, and the kernel is the set of all fractional ideals of the form $(\frac{a}{b})$, where a and b are positive integers such that $(a, m) = 1, (b, m) = 1$ and $a \equiv b \mod m$.

Proof. See [[Janusz], Chapter III, Section 3, Proposition 3.3, (p. 133)]. □

4.4 Analytic methods

Chevelley's algebraic approach to class field theory, often regarded as the modern approach uses idéles and the machinery of cohomology of groups. But, we consider the historically older approach of using Dirichlet series and infinite series to prove results about the distribution of prime ideals. The main result – the surjectivity of the Artin map is established without the need of heavy machinery and is done via the Frobenius density theorem, which, roughly speaking, says that in any Galois extension of number fields, infinitely many prime ideals have the same Frobenius automorphism.

Let K be a number field, $\mathcal{O}_K$ denote its ring of algebraic integers, $\mathcal{I}_K$, the group of non-zero fractional ideals of $\mathcal{O}_K$, and $K^\times$ its multiplicative group. Let $\mathrm{Cl}(K)$ denote the ideal class group of K.

Let $\iota : K^\times \to \mathcal{I}_K$ be the natural map defined by $\iota(\alpha) = (\alpha) \in \mathcal{I}_K$. The kernel of ι is precisely the unit group U_K of $\mathcal{O}_K$. The structure of U_K is given by Theorem 3.0.1. We also get the following exact sequence

$$1 \longrightarrow U_K \longrightarrow K^\times \longrightarrow \mathcal{I}_K \longrightarrow \mathrm{Cl}(K) \longrightarrow 1 \tag{4.2}$$

As mentioned before, class field theory gives an explicit description of the abelian extensions of number fields in terms of the arithmetic of the base number field. By the arithmetic of the base number field, we mean the subgroups of the groups occurring in Equation (4.2).

4.4.1 The ray class group

A reference for this section is Milne's notes on class field theory [[Milne-CFT], Chapter V, Section 1, (p. 147)].

Definition 4.4.1 (Modulus). A modulus $\mathfrak{m}$ for K is defined as a formal product

$$\mathfrak{m} = \prod_{\mathfrak{p}} \mathfrak{p}^{n(\mathfrak{p})}$$

taken over all infinite and finite primes $\mathfrak{p}$ of K such that each exponent $n(\mathfrak{p})$ is a non-negative integer and $n(\mathfrak{p}) = 0$ for almost all primes $\mathfrak{p}$. Furthermore, if $\mathfrak{p}$ is a complex infinite prime, then $n(\mathfrak{p}) = 0$ and if $\mathfrak{p}$ is a real infinite prime, then $n(\mathfrak{p}) = 0$.

Note 4.4.2. For any number field K and a modulus $\mathfrak{m}$ for K, we can write $\mathfrak{m}$ as a product $\mathfrak{m} = \mathfrak{m}_0\mathfrak{m}_\infty$, where

$$\mathfrak{m}_0 := \prod_{\mathfrak{p}\text{ finite}} \mathfrak{p}^{n(\mathfrak{p})} \quad \text{and} \quad \mathfrak{m}_\infty := \prod_{\mathfrak{p}\text{ real}} \mathfrak{p}^{n(\mathfrak{p})}$$

We call the integral ideal $\mathfrak{m}_0$ as the *finite part* of $\mathfrak{m}$ and $\mathfrak{m}_\infty$ as the *infinite part* of $\mathfrak{m}$.

We say that a modulus $\mathfrak{m}_1 = \prod_{\mathfrak{p}} \mathfrak{p}^{n_1(\mathfrak{p})}$ for K divides a modulus $\mathfrak{m}_2 = \prod_{\mathfrak{p}} \mathfrak{p}^{n_2(\mathfrak{p})}$ for K if $n_1(\mathfrak{p}) \leq n_2(\mathfrak{p})$ for all primes $\mathfrak{p}$. In particular, we note that a prime $\mathfrak{p}$ divides a modulus $\mathfrak{m}$ if and only if $n(\mathfrak{p}) > 0$.

Let us recall that given a number field K and a finite prime $\mathfrak{p}$ of K, we get a discrete valuation $v_\mathfrak{p}$ on K given by $v_\mathfrak{p}(a) = \mathrm{ord}_\mathfrak{p}(a) = m$, where $a = u\pi^m$ for a fixed uniformizer $\pi \in O_K$, some unit $u \in O_K^\times$, and some integer $m \in \mathbb{Z}$.

We say that a fractional ideal $I \in \mathcal{I}_K$ is co-prime (or prime) to $\mathfrak{m}$ if $v_\mathfrak{p}(I) = 0$ for all finite primes $\mathfrak{p}$ dividing $\mathfrak{m}_0$. Let $\mathcal{I}_K^\mathfrak{m}$ denote the subgroup of the fractional ideal group $\mathcal{I}_K$ of K consisting of all fractional ideals co-prime to $\mathfrak{m}$.

For a given modulus $\mathfrak{m} = \prod_{\mathfrak{p}} \mathfrak{p}^{n(\mathfrak{p})}$ for K, we define subgroups $K_\mathfrak{m}$ and $K_{\mathfrak{m},1}$ of $K^\times$ as follows:

$$K_\mathfrak{m} := \left\{ a \in K^\times \;\middle|\; (a) \in \mathcal{I}_K^\mathfrak{m} \right\}$$

$$K_{\mathfrak{m},1} := \left\{ a \in K^\times \;\middle|\; \begin{array}{c} v_\mathfrak{p}(a-1) \geq n(p), \quad \text{for all finite } \mathfrak{p} \text{ dividing } \mathfrak{m} \\ a_\mathfrak{p} > 0, \quad \text{for all real } \mathfrak{p} \text{ dividing } \mathfrak{m} \end{array} \right\}$$

We note that $K_{\mathfrak{m},1} \leqslant K_{\mathfrak{m}}$ is the subgroup consisting of all elements $a \in K_{\mathfrak{m}}$ with $v_{\mathfrak{p}}(a-1) \geq v_{\mathfrak{p}}(\mathfrak{m}_0)$ for all finite primes $\mathfrak{p}|\mathfrak{m}_0$ and $a_{\mathfrak{p}} > 0$ for all infinite (real) primes $\mathfrak{p}$ dividing $\mathfrak{m}_\infty$.

The group $\mathscr{R}_K^{\mathfrak{m}}$ of principal fractional ideals (a) such that $(a) \in \mathscr{I}_K^{\mathfrak{m}}$ and $a \in K_{\mathfrak{m},1}$ is called a *ray* or a *ray group*[1].

Definition 4.4.3 (Ray class group (modulo $\mathfrak{m}$)). Let $\mathfrak{m}$ be a modulus for a number field K. The *ray class group (modulo* $\mathfrak{m}$*)* $\mathrm{Cl}_K^{\mathfrak{m}}$ for the modulus $\mathfrak{m}$ is defined as the quotient

$$\mathrm{Cl}_K^{\mathfrak{m}} = \mathscr{I}_K^{\mathfrak{m}} / \mathscr{R}_K^{\mathfrak{m}}$$

An interesting result about ray class groups is that for any modulus $\mathfrak{m}$ for a number field K, the ray class group $\mathscr{I}_K^{\mathfrak{m}}/\mathscr{R}_K^{\mathfrak{m}}$ is a finite group. (See [[Janusz], Chapter IV, Section 1, Corollary 1.6, (p. 141)])

Remark 4.4.4. We shall denote the order of the ray class group $\mathrm{Cl}_K^{\mathfrak{m}}$ by $h_{\mathfrak{m}}$. If we choose the modulus $\mathfrak{m}$ to be an empty product, i.e., $\mathfrak{m} = 1$, then the ray class group $\mathrm{Cl}_K^{\mathfrak{m}}$ is just the ordinary ideal class group Cl_K, the latter's order which, we denote by h_K. We now make a nice observation that for any number field K, its class number h_K divides the order of the ray class group $h_{\mathfrak{m}}$ for any modulus $\mathfrak{m}$ for K.

4.4.2 Zeta functions of number fields

Let K be a number field and let O_K denote its ring of integers. For any integral ideal $\mathfrak{U}$ of O_K, we recall that the counting norm $\mathscr{N}(\mathfrak{U})$ is defined as the unique positive generator of the $\mathbb{Z}-$ideal $N_{K/\mathbb{Q}}(\mathfrak{U})$, which is also equal to the cardinality of the finite ring $O_K/\mathfrak{U}$.

For any positive integer n, we know that there exists only a finite number $a_K(n)$ of integral ideals of norm n. We now define the ζ-function of K as

$$\zeta_K(s) = \sum_{n=1}^{\infty} \frac{a_K(n)}{n^s} = \sum_{\mathfrak{U} \text{ integral} \neq 0} \frac{1}{\mathscr{N}(\mathfrak{U})^s}$$

Remark 4.4.5. The Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$ is just the zeta function of the rational number field.

We now mention a few important results concerning zeta functions of number fields.

Proposition 4.4.6.

[1] Janusz in [Janusz] denotes ray groups $\mathscr{R}_K^{\mathfrak{m}}$ by $\iota(K_{\mathfrak{m},1})$ to denote them as the image of the group $K_{\mathfrak{m},1}$ under the map $\iota : K^\times \to \mathscr{I}_K$ mapping $a \mapsto (a) \in \mathscr{I}_K$

1.
$$\zeta_K(s) = \prod_{\substack{\mathfrak{p} \\ \text{prime ideals } \mathfrak{p} \text{ in } O_K}} (1 - \mathcal{N}(\mathfrak{p})^{-s})^{-1}$$

2.
$$\zeta_{\mathbb{Q}}(s) = \prod_{\substack{p \\ \text{positive prime integers } p}} (1 - p^{-s})^{-1}$$

Proof. See [[Janusz], Chapter IV, Section 4, Corollary 4.3, (p. 157)] □

Definition 4.4.7. For any two complex analytic functions $f_1(s)$ and $f_2(s)$ defined (at least) on the upper half complex plane $\mathrm{Re}(s) > 1$, we write $f_1(s) \sim f_2(s)$ if $f_1(s) - f_2(s)$ has a finite limit as s approaches 1 in the common domains.

Theorem 4.4.8. Let K be a number field. Let S denote the set of all prime ideals of O_K which have relative degree one over $\mathbb{Q}$. (The relative degree $f(\mathfrak{p}|p)$ of a prime ideal $\mathfrak{p}$ of O_K is defined as the degree $[O_K/\mathfrak{p} : \mathbb{Z}/(p)]$ of the field extension $O_K/\mathfrak{p}$ / $\mathbb{Z}/(p)$).

Then,
$$\log(s-1) \sim \sum_{\mathfrak{p} \in S} \frac{1}{\mathcal{N}(\mathfrak{p})^s}$$

Thus, as $\log(s-1)$ is not bounded at $s = 1$, the sum on the right must have infinitely many terms. Hence, there are *infinitely many* prime ideals of O_K that have relative degree one over $\mathbb{Q}$.

Proof. See [[Janusz], Chapter IV, Section 4, Theorem 4.6, (p. 159)]. □

4.4.3 Dirichlet Density

Definition 4.4.9 (Dirichlet density). Let K be a number field. Let S be a set of some prime ideals in O_K. We say that the set S has a *Dirichlet density* of δ and write $\delta(S) = \delta$ if there exists a real number δ such that
$$-\delta \log(s-1) \sim \sum_{\mathfrak{p} \in S} \frac{1}{\mathcal{N}(\mathfrak{p})^s}$$

Note 4.4.10. If S is the set of all prime ideals of the number ring O_K of a number field K which have relative degree one over $\mathbb{Q}$, then from Theorem 4.4.8 we see that $\delta(S) = 1$.

We list below an important property of Dirichlet density.

Remark 4.4.11. If S is a set of finite primes of a number ring O_K that has Dirichlet density $\delta(S) \neq 0$, then S is an infinite set.

Proof. We prove that a finite set S of prime ideals of O_K has a zero Dirichlet density. This is because, if S is a finite set, then

$$\sum_{\mathfrak{p}\in S} \frac{1}{\mathcal{N}(\mathfrak{p})^s} \sim 0.$$

□

Our general theme to show the existence of certain kinds of prime ideals or prove their infinitude would be to prove that the set of all such prime ideals has a non-zero Dirichlet density.

THEOREM 4.4.12 (Density of finite primes in a subgroup of the ray class group). Let $\mathfrak{m}$ be a modulus for a number field K and let H be a group such that $\mathscr{R}_K^{\mathfrak{m}} \subset H \subset \mathscr{I}_K^{\mathfrak{m}}$. Let h be the index $[\mathscr{I}_K^{\mathfrak{m}} : H]$ of H in $\mathscr{I}_K^{\mathfrak{m}}$. Let S be the set of prime ideals contained in the subgroup H. If S has a Dirichlet density, then $\delta(S) \leq 1/h$.

Proof. See [[Janusz], Chapter IV, Section 4, Theorem 4.8, (p. 161)]. □

4.4.4 The Frobenius Density Theorem

We first recall that for any number field K, and a set S containing the finite primes of O_K, we use the notation $\mathscr{I}_K^S$ to denote the subgroup $\mathscr{I}_K^S$ of $\mathscr{I}_K$ generated by the non-zero prime ideals which are away from S, i.e., which do not lie in S. The main application of the Frobenius density theorem is to prove that the Artin map $\phi_{L/K} : \mathscr{I}_K^S \to G$ is surjective, where, S contains all the primes of K that ramify in L.

DEFINITION 4.4.13 (Division of σ). Let σ be an element of order n in a group G. Then, *the division of* σ is the collection of all elements of G that are conjugate to some element of the form σ^m, for some integer m co-prime to n.

NOTE 4.4.14. Let σ be an element of order n in G, let H be the cyclic group $\langle\sigma\rangle$, and t be the number of elements in the division of σ. Then,

$$t = \phi(n)\,[G : N_G(H)],$$

where $\phi(n)$ is the Euler's totient function and $N_G(H)$ is the normalizer of H in G.

Proof. See [[Janusz], Chapter IV, Section 5, Proposition 5.1, (p. 162)]. □

THEOREM 4.4.15 (Frobenius density theorem). Let L/K be a Galois extension of number fields. Let $G = \mathrm{Gal}(L/K)$. Let t be the number of elements in the division of some element σ in G. Let S_1 be the set of all primes of K which are divisible by a prime of L which has its Frobenius automorphism in the division of σ. Then, S_1 has a Dirichlet density of $t/|G|$.

Proof. See [[Janusz], Chapter IV, Section 5, Theorem 5.2, (p. 162)]. □

The main applications of the analytic methods we have discussed in this section are to prove the surjectivity of the Artin map.

Surjectivity of the Artin map

THEOREM 4.4.16. Let L/K be an abelian extension of number fields. Let S be the set of prime ideals of O_K that ramify in L. Then, the Artin map $\phi_{L/K}: \mathcal{I}_K^S \to \mathrm{Gal}(L/K)$ is surjective.

Proof. Let $\sigma \in \mathrm{Gal}(L/K)$ be arbitrary. As L/K is abelian, the division of σ consists of the elements in the cyclic group $\langle\sigma\rangle$. Now, using the Frobenius density theorem (Theorem 4.4.15) we get that there exist infinitely many primes of L whose Frobenius automorphism generates $\langle\sigma\rangle$. As S is finite, we now choose a prime $\mathfrak{P}$ of L such that $\mathfrak{p} := \mathfrak{P} \cap O_K$ is not in S. Then, the Frobenius automorphism $\phi_{L/K}(\mathfrak{p})$ generates $\langle\sigma\rangle$ and hence, $\sigma \in \phi_{L/K}(\mathcal{I}_K^S)$. As σ was an arbitrary element of $\mathrm{Gal}(L/K)$, we conclude that $\phi_{L/K}$ is surjective. □

Few more applications of the Frobenius density theorem

Using the notion of Dirichlet density, we were able to establish results such as the infinitude of the (finite) primes of any number field that have relative degree one over $\mathbb{Q}$. We note that we can answer a few more similar questions by getting them as simple corollaries to the Frobenius density theorem.

COROLLARY 4.4.17 (Corollaries to Theorem 4.4.15).

1. Assume that L/K is a cyclic extension of order n of number fields. Let d be any positive divisor of n. Let $\mathrm{Spl}(L/K)$ denotes the set of primes in K which split completely in L. Let $S(d)$ denote the set of finite primes of O_K which have exactly d many prime factors in O_L. Then, $S(d)$ has a non-zero Dirichlet density $\phi(n/d)/n$. In particular, *there exist infinitely many prime ideals of O_K which remain prime in O_L.*

   Proof. Ignoring the finitely many primes of K which ramify in L, we note that a prime $\mathfrak{p}$ of O_K has d factors in O_L if and only if $\sigma := \phi_{L/K}(\mathfrak{p})$ has order an order of n/d. Now, using the Frobenius density theorem (Theorem 4.4.15), we observe that $\delta(S(d)) = \phi(n/d)/n$. □

   REMARK 4.4.18. Quite interestingly, whenever L/K is a non-cyclic Galois extension of (non-trivial) number fields, no prime of K remains inert in L! (see [MSE-2])

2. (Bauer's Theorem – 1903). Let L_1 and L_2 be two (finite) Galois extensions of a number field K. Let S_i denote the set of prime ideals in $\mathcal{O}_K$ which *split completely* in L_i, for $i = 1, 2$. $S_1 \subset S_2$ *(except possibly for a set of zero Dirichlet density – i.e., except possibly whenever* $\delta(S_1 \setminus S_2) = 0$*)* if and only if $L_2 \subset L_1$.

   In particular, $L_1 = L_2$ if and only if $\mathrm{Spl}(L_1/K) = \mathrm{Spl}(L_2/K)$.

   Proof. $\boxed{\Leftarrow}$ If $L_2 \subset L_1$, then any finite prime of K that splits completely in L_1 also splits completely in L_2.

   $\boxed{\Rightarrow}$ We now suppose that $S_1 \subset S_2$ except possibly for a set of density zero. We now consider the composite extension $L = L_1L_2$. We note that as a result of the relationship between an unramified prime which splits completely and its corresponding Frobenius automorphism (see [[Janusz], Chapter III, Section 2, Properties 2.4 and 2.5, (p. 127-128)]), the primes that split completely in L are precisely those that split completely in both L_1 and L_2. Using the Frobenius density theorem (Theorem 4.4.15), we compute the (Dirichlet) densities of S_1 and S_2 to get
   $[L : K]^{-1} = \delta(S_1) = [L_1 : K]^{-1}$. Hence, $L_1L_2 = L = L_1$ which implies that $L_2 \subset L_1$. □

   Remark 4.4.19. The above result is an example of a classification of all normal extensions of a number field K in terms of the objects in K alone – namely, in this case, certain set of primes of K. However, the solution to the problem of characterizing those primes that arise as the primes that split completely in some (finite) normal extension of K is only known (via class field theory) for abelian extensions of K. Moreover, Bauer's above theorem doesn't give us a rule for describing the set of primes that split completely in a Galois extension of number fields L/K. The latter is however, known for abelian extensions L/K via class field theory in terms of generalized congruences. Fore more details, see [[Conrad-CFT-History], Section 2, (p. 4)].

The Artin Reciprocity Theorem

We have seen that the Artin map $\phi_{L/K}$ carries $\mathcal{I}_K^{\mathfrak{m}}$ onto $\mathrm{Gal}(L/K)$, for a Galois extension of number fields L/K, and whenever the modulus $\mathfrak{m}$ contains the finite primes of K that ramify in L. Thus, $\mathcal{I}_K^{\mathfrak{m}} \cap \ker \phi_{L/K}$ has index $[L : K]$ in $\mathcal{I}_K^{\mathfrak{m}}$.

For a cyclic extension L/K of number fields, we note that the group $N_{L/K}(\mathcal{I}_L^{\mathfrak{m}}) \cdot \mathcal{R}_K^{\mathfrak{m}}$ has index $[L : K]$ in $\mathcal{I}_K^{\mathfrak{m}}$ for a suitable modulus $\mathfrak{m}$. (For the proof using the cohomology of

cyclic groups, we refer to [[Janusz], Chapter V, Section 4, Theorem 4.4, (p. 188)]).

The Artin reciprocity theorem says under the restriction that L/K is an abelian extension of number fields, and for a suitable modulus $\mathfrak{m}$, the two subgroups $\mathcal{I}_K^{\mathfrak{m}} \cap \ker \phi_{L/K}$ and $N_{L/K}(\mathcal{I}_L^{\mathfrak{m}}) \cdot \mathcal{R}_K^{\mathfrak{m}}$ are in fact, equal. We restate the same as follows:

THEOREM 4.4.20 (Artin reciprocity). Let L/K be an abelian extension of number fields. Let $\mathfrak{m}$ be a modulus for K which contains (divides) all the finite primes of K that ramify in L. If the exponents of the prime divisors of $\mathfrak{m}$ are sufficiently large, then the Artin map $\phi_{L/K}$ from $\mathcal{I}_K^{\mathfrak{m}}$ to $\mathrm{Gal}(L/K)$ is surjective and the kernel of this map is $N_{L/K}(\mathcal{I}_L^{\mathfrak{m}}) \cdot \mathcal{R}_K^{\mathfrak{m}}$.

Proof. See [[Janusz], Chapter V, Section 5, Theorem 5.8, (p. 197)]. □

REMARK 4.4.21. We note that the Kronecker-Weber theorem can be derived as a corollary to the Artin reciprocity theorem. For the proof, we refer to [[Janusz], Chapter V, Section 5,Theorem 5.10, (p. 198)].

4.4.5 The Hilbert Class Field

For the proofs in this section, we refer to [Janusz] and [Milne-CFT].

Let K be a number field. Then, there exists a one-to-one correspondence between (finite) unramified abelian extensions N of K and the subgroups H of the ideal class group Cl_K. Furthermore, if the extension N/K corresponds to the subgroup $H \subset \mathrm{Cl}_K$, then the Artin map induces an isomorphism,

$$\mathrm{Cl}(K)/H \xrightarrow{\sim} \mathrm{Gal}(M/K) \tag{4.3}$$

Thus, there is a unique unramified abelian extension L of K such that the Artin map $\phi_{L/K}$ induces an isomorphism

$$\mathrm{Cl}(K) \xrightarrow{\sim} \mathrm{Gal}(L/K)$$

The field L is called the *Hilbert class field* of K and is characterized by the following properties:

1. L is a finite abelian unramified extension of K.
2. If L' is a finite abelian unramified extension of K, then $L' \subset L$.

NOTE 4.4.22. The Hilbert class field L of K is thus known as the maximal unramified abelian extension of K.

As a consequence, we also get the characterization of the primes of K which split completely in the Hilbert class field L of K. Namely, a prime $\mathfrak{p}$ of K splits completely in L if and only if $\mathfrak{p}$ is a principal ideal.

REMARK 4.4.23. The theory of Hilbert class fields can be used to answer questions about primes p of the form $p = x^2 + ny^2$, for infinitely many integers n. For more details, we refer the reader to [[Cox], Chapter Two, §5, (p. 86)].

Chapter 5

Some computations and examples

5.1 Quadratic number fields

5.1.1 Discriminant and Integral bases of Quadratic number fields

A Quadratic number field is a degree 2 extension of $\mathbb{Q}$. Using the *quadratic formula*[1], it is not hard to show that any quadratic number field is an abelian extension of the form $\mathbb{Q}(\sqrt{D})$, for a square-free integer $D \in \mathbb{Z}$.

We recall that the discriminant d_K of a number field K is given by

$$d_K = \det(Tr_{K/\mathbb{Q}}([\omega_i \cdot \omega_j]_{ij})) = \det([\omega_i^{(j)}]_{ij})^2 \tag{5.1}$$

where $\omega_1, \ldots, \omega_n$ is an integral basis for K and $\omega_i^{(j)}$ denotes the jth conjugate of ω over $\mathbb{Q}$.

An integral basis $\mathcal{B}$ for K consists of the following: (See [[Murthy-Esmonde] Chapter 4, Section 4.3, Example 4.3.4, p. 47])

$$\text{When } D \equiv 1 \mod 4,\ \mathcal{B} = \{1, \tfrac{1+\sqrt{D}}{2}\},\ \text{and}$$
$$\text{When } D \equiv 2, 3 \mod 4,\ \mathcal{B} = \{1, \sqrt{D}\}$$

As we have an explicit description of the integral basis depending on the square free integer D, we compute the discriminant d_K for a quadratic number field using Equation (5.1), which is given by the following formula:

$$d_K = \begin{cases} (-1)^{r_2} \cdot D & \text{if } D \equiv 1 \mod 4 \\ (-1)^{r_2} \cdot 4D & \text{if } D \equiv 2, 3 \mod 4 \end{cases}$$

[1]The complex roots of a quadratic equation $aX^2 + bX + c \in \mathbb{Z}[X]$ are given by the quadratic formula as $X_{1,2} = \frac{-b}{a} \pm \frac{\sqrt{b^2-4ac}}{2a}$

where, r_2 denotes the number of distinct pairs of non-real embeddings of K in $\mathbb{C}$.

We also note that the ring of integers O_K is given by

$$O_K = \begin{cases} \mathbb{Z}[\frac{1+\sqrt{D}}{2}] & \text{if } D \equiv 1 \mod 4 \\ \mathbb{Z}[\sqrt{D}] & \text{if } D \equiv 2, 3 \mod 4 \end{cases}$$

We further have the following simplified formulas for the Minkowski bound $M = \frac{n!}{n^n} \cdot \left(\frac{4}{\pi}\right)^{r_2} \cdot \sqrt{|d_K|}$ for abelian degree 2 extensions of $\mathbb{Q}$.

1. When *K is a real quadratic extension:* $M = \frac{1}{2} \cdot \sqrt{d_K}$
2. When *K is an imaginary quadratic extension:* $M = \frac{2}{\pi} \cdot \sqrt{d_K}$

5.2 Class groups of Quadratic number fields with small Minkowski bounds

We compute the class groups of some quadratic number fields with small Minkowski bounds using the method described in Section 3.1.1.

5.2.1 Example 1: $\mathbb{Q}(\sqrt{-6})$

Let $K := \mathbb{Q}(\sqrt{-6})$. As $6 \equiv -1 \cdot 2 \equiv 2 \mod 4$, $\{1, \sqrt{-6}\}$ is an integral basis for $\mathbb{Q}(\sqrt{-6})$. The number ring of $\mathbb{Q}(\sqrt{-6})$ is $\mathbb{Z}[\sqrt{-6}]$. The discriminant (over $\mathbb{Q}$) of $\mathbb{Q}(\sqrt{-6})$ is $d_K = 24$ The Minkowski bound M in this case is $M \approx 3.12$. Using Theorem 3.1.3 and Section 3.1.1, the class group $\mathrm{Cl}(K)$ is generated by the prime ideals with norm less than or equal to 3. As the polynomial $X^2 + 6$ is reducible modulo 2, the factorization of the ideal (2) in O_K is $\mathfrak{p}_2^2$, for the unique prime $\mathfrak{p}_2$ which lies over 2. Thus, $\mathrm{Cl}(K) = \langle \mathfrak{p}_2 \rangle$. We note that for any algebraic integer $a + b\sqrt{-6} \in O_K$, the norm $N_{L/K}(a + \sqrt{-6}b) = a^2 + 6b^2$. Hence, the prime ideal $\mathfrak{p}_2$ is not principal as the equation $x^2 + 6y^2 = 2$ doesn't have any positive integer solutions in x and y. Further, as $\mathfrak{p}_2^2 = (2) \sim (1)$, we conclude that $\mathrm{Cl}(K) \cong \mathbb{Z}/(2)$. △

5.2.2 Example 2: $\mathbb{Q}(\sqrt{-10})$

Let $K := \mathbb{Q}(\sqrt{-10})$. We note that $-10 \equiv 2 \mod 4$, and hence, $d_K = (-1)^{r_2} \cdot 4D = 40$. The Minkowski bound $M \approx 4.03$. The class group of K is thus generated by the prime ideals which lie above 2 and 3. We also note that $O_K = \mathbb{Z}[\sqrt{D}]$.

Table 5.1: Factorization of some ideals in O_K, where $K = \mathbb{Q}(\sqrt{-10})$

prime number p	Irreducible factors of $(X^2 + 10)$ mod p	(p) in O_K
2	X^2	$\mathfrak{p}_2^2$
3	$X^2 + 1$ *Irreducible*	*prime*

Therefore, the class group $\mathrm{Cl}(K)$ is generated by $[\mathfrak{p}_2]$ where, $[\mathfrak{p}_2]^2 = \mathbb{1}$. Moreover, similar to Section 5.2.1, as the equation $x^2 + 10y^2$ doesn't have any positive integer solutions in x and y, we conclude that the prime ideal $\mathfrak{p}_2$ is not principal. Thus, $\mathrm{Cl}(K) \cong \mathbb{Z}/(2)$. △

5.2.3 Example 3: $\mathbb{Q}(\sqrt{-35})$

Let $K := \mathbb{Q}(\sqrt{-35})$. As $35 \equiv (-1)\cdot(-1) \equiv 1 \mod 4$, the discriminant $d_K = (-1)^{r_2} \cdot D = 35$.

The ring of integers $O_K = \mathbb{Z}[\frac{1+\sqrt{D}}{2}]$. An integral basis for K is 1, $\frac{1+\sqrt{D}}{2}$ and the minimal polynomial of $\frac{1+\sqrt{D}}{2}$ over $\mathbb{Q}$ is $X^2 - 2X + 9$. The Minkowski bound $M = \frac{2}{\pi} \cdot \sqrt{d_K} \approx 3.77$. The class group is generated by the prime ideals $\mathfrak{p}_2$, $\mathfrak{p}_3$, and $\mathfrak{p}_3'$ which lie over the rational primes 2, 3 and 3 respectively. We shall also compute the prime factorization of the ideals (5), (7) and (13) as they shall help us later.

Table 5.3: Factorization of some ideals in O_K, where $K = \mathbb{Q}(\sqrt{-35})$

prime number p	Irreducible factors of $(X^2 - 2X + 9)$ mod p	(p) in O_K
2	$(X + 1)^2$	$\mathfrak{p}_2^2$
3	$X(X + 1)$	$\mathfrak{p}_3 \cdot \mathfrak{p}_3'$
5	*Irreducible*	*prime*
7	*Irreducible*	*prime*
13	*Irreducible*	*prime*

We also make a few norm computations: Let $\alpha := \frac{1+\sqrt{D}}{2}$

$$N_{K/Q}(\alpha+1) = 2^2 \cdot 3^2 \qquad N_{K/\mathbb{Q}}(\alpha+2) = 3 \cdot 13$$
$$N_{K/Q}(\alpha+5) = 2^2 \cdot 3 \cdot 5$$

Suppose that $\mathfrak{p}_3$ is the prime which divides the ideal $(\alpha+2)$. Hence, $\mathfrak{p}_3$ is principal. Moreover, as $\mathfrak{p}_3 \cdot \mathfrak{p}_3' = (3)$, we conclude that $\mathfrak{p}_3'$ is principal as well. Thus, $\mathrm{Cl}(K) = \langle[\mathfrak{p}_2]\rangle$. We further know that $[\mathfrak{p}_2]$ is 2−torsion as $\mathfrak{p}_2^2 = (2)$. We claim that $\mathfrak{p}_2$ is not principal, and hence, the class group is of order 2.

Proof of our claim: We first note that as a result of [[Murthy-Esmonde], Chapter 4, Exercise 4.3.5, (p. 47)], any element of $\mathcal{O}_K$ can be written of the form $\frac{a}{2} + \frac{b\sqrt{D}}{2}$, for integers a, b such that $a \equiv b \mod 2$. We now compute the norm of a general algebraic element. We see that $N_{K/\mathbb{Q}}(\frac{a}{2} + \frac{b\sqrt{D}}{2}) = \det\left(\begin{bmatrix} \frac{a}{2} & \frac{-35b}{2} \\ \frac{b}{2} & \frac{a}{2} \end{bmatrix}\right) = \frac{a^2+35b^2}{4}$.

Thus, if $\mathfrak{p}_2$ were to be principal, then, $2 = \frac{x^2+35y^2}{4}$ has positive integer solutions in x and y with $x \equiv y \mod 2$ if and only if $8 = x^2 + 35y^2$ has positive integer solutions in x and y with $x \equiv y \mod 2$, the latter which, is absurd.

Thus, $\mathrm{Cl}(K) \cong \mathbb{Z}/(2)$. △

5.2.4 Example 4: $\mathbb{Q}(\sqrt{-14})$

We first note that $-14 \equiv 2 \mod 4$ and hence the discriminant d_K of $K := \mathbb{Q}(\sqrt{-14})$ is $d_K = (-1)^{r_2} \cdot 4D = 56$. The number ring $\mathcal{O}_K = \mathbb{Z}[\sqrt{-14}]$. The Minkowski bound $M = \frac{2}{\pi} \cdot \sqrt{d_K} \approx 4.76$. Hence, we consider the factorization of the ideals (2) and (3) in $\mathcal{O}_K$. We observe the following:

Table 5.6: Factorization of some ideals in $\mathcal{O}_K$, where $K = \mathbb{Q}(\sqrt{-14})$

prime number p	Irreducible factors of $(X^2+14) \mod p$	(p) in $\mathcal{O}_K$
2	X^2	$\mathfrak{p}_2^2$
3	$(X+1)(X+2)$	$\mathfrak{p}_3 \cdot \mathfrak{p}_3'$

We hence have that $[\mathfrak{p}_2]^2 = \mathbb{1}$ and $[\mathfrak{p}_3'] = [\mathfrak{p}_3]^{-1}$. Therefore, $\mathrm{Cl}(K) = \langle[\mathfrak{p}_2], [\mathfrak{p}_3]\rangle$.

Since $N_{K/\mathbb{Q}}(a + \sqrt{-14}b) = a^2 + 14b^2$, we see that the prime ideals $\mathfrak{p}_2$, $\mathfrak{p}_3$ and $\mathfrak{p}_3'$ are all not principal. Let $\alpha := \sqrt{-14}$. We observe that:

$$N_{K/Q}(\alpha) = 2 \cdot 7 \qquad N_{K/\mathbb{Q}}(\alpha + 1) = 3 \cdot 5$$
$$N_{K/Q}(\alpha + 2) = 2 \cdot 3^2 \qquad N_{K/\mathbb{Q}}(\alpha + 4) = 2 \cdot 3 \cdot 5$$

Hence, $(\alpha) = \mathfrak{p}_2 \cdot \mathfrak{p}_7$ and $(\alpha + 1) = \mathfrak{p}_3 \cdot \mathfrak{p}_5$, say, for prime ideals $\mathfrak{p}_2, \mathfrak{p}_3, \mathfrak{p}_5, \mathfrak{p}_7$ lying over $2, 3, 5, 7$ respectively. Also, $\mathfrak{p}_3 \nmid (\alpha + 2)$. (This is because, if $\mathfrak{p}_3 \mid (\alpha + 1)$ and $\mathfrak{p}_3 \mid (\alpha + 2)$, then $\mathfrak{p}_3$ would divide the difference $(\alpha + 2 - \alpha - 1) = (1)$, which is absurd.)

Thus, $(\alpha + 2) = \mathfrak{p}_2 \cdot (\mathfrak{p}_3')^2 \Rightarrow [\mathfrak{p}_3']^2 = [\mathfrak{p}_3]^{-2} = [\mathfrak{p}_2]^{-1}$. Therefore, $[\mathfrak{p}_2] = [\mathfrak{p}_3]^2$.

Hence, $\mathrm{Cl}(K) = \langle [\mathfrak{p}_3] \rangle$, and we are left with the task of finding the order of $[\mathfrak{p}_3]$.

We now note the following:

1. $[\mathfrak{p}_5] = [\mathfrak{p}_5'] = [\mathfrak{p}_3]^{-1}$, where $[\mathfrak{p}_5']$ is the prime ideal class containing the image of $\alpha + 4$.
2. $[\mathfrak{p}_5] = [\mathfrak{p}_3]^{-1}$

Hence, $\mathbb{1} = [\mathfrak{p}_2] \cdot [\mathfrak{p}_3] \cdot [\mathfrak{p}_5'] = [\mathfrak{p}_3]^2 \cdot [\mathfrak{p}_3] \cdot [\mathfrak{p}_3]$ and $[\mathfrak{p}_3]$ has order dividing 4. Further, as $[\mathfrak{p}_2] = [\mathfrak{p}_3]^2$ is not trivial ($\mathfrak{p}_2$ was shown to be not principal), we conclude that the order of $[\mathfrak{p}_3]$ is 4.

Hence, $\mathrm{Cl}(K) \cong \mathbb{Z}/(4)$. △

5.3 A method to find the integral basis of degree 3 abelian extensions of $\mathbb{Q}$

We first state the following result under some assumptions, which gives us a number ring which is free on a single generator.

Proposition 5.3.1. For any number field K, if $\exists\, \theta \in O_K$ such that $d_{K/\mathbb{Q}}(\theta)$ is square-free, then $O_K = \mathbb{Z}[\theta]$.

Proof. See [[Murthy-Esmonde], Chapter 4, Section 4.5, Exercise 4.5.26, (p. 225)]. □

We list another scenario when the ring of integers of a number field is free on a single generator.

Proposition 5.3.2. Suppose the minimal polynomial $f(X) \in \mathbb{Z}[X]$ of an algebraic integer $\theta \in \mathbb{C}$ is Eisenstenian with respect to a prime number p. Let $K := \mathbb{Q}(\theta)$. Then, $O_K = \mathbb{Z}[\theta]$.

Proof. See [[Murthy-Esmonde], Chapter 4, Section 4.5, Exercise 4.5.3, (p. 214)]. □

As we usually don't encounter number fields with square-free discriminants or with their primitives being roots of Eisenstenian polynomials often[2], we outline a method to find the integral basis of number fields. This method is much simpler for degree 3 abelian extensions of $\mathbb{Q}$ as the computations are not very tedious to be done by hand.

Remark 5.3.3. A general method to find the integral basis of any number field called the *Zassenhaus's Round* 2 *Algorithm* is known which is very tedious to be done by hand. We refer to [[Cohen], Chapter 6, Section 6.1.4, Algorithm 6.1.8] for more details.

5.3.1 Normalizing the cubic equation

Consider a degree 3 monic irreducible polynomial $f(X)$ over $\mathbb{Q}$, where $f(X)$ is the minimal polynomial of $\alpha \in K$ such that $K = \mathbb{Q}(\alpha)$ is a number field. Suppose $f(X) = X^3 + aX^2 + bX + c$. We make the substitution $X = Y - a/3$ to get $g(Y) = Y^3 + pY + q$, where $p = \frac{1}{3} \cdot (3b - a^2)$ and $q = \frac{1}{27} \cdot (2a^3 - 9ab + 27c)$. This substitution for a general cubic equation is known as *normalizing the cubic*.

As the roots of the polynomials $f(X)$ and $g(Y)$ differ by a constant rational number $a/3$, we note that the discriminants $\Delta(f)$ and $\Delta(g)$ are both equal. i.e., $\Delta(f) = \Delta(g) = -4p^3 - 27q^2$. This is because the formula for the discriminant of polynomials involves the differences of its roots. Also, their corresponding splitting fields $\mathrm{Split}(f; \mathbb{Q})$ and $\mathrm{Split}(g; \mathbb{Q})$ are equal. (See [[Dummit-Foote], Chapter 14, Section 14.6, (p. 612)] for more details.)

Some of the advantages of normalizing the cubic are as follows:

1. We may end up with $p = 0$ to get a pure cubic field upon normalizing, in which case, the ring of integers is known completely, as we shall describe later.
2. We may also find that the normalized cubic is Eisenstenian with respect to some prime, in which case we can use Proposition 5.3.2.

Theorem 5.3.4 (Integral basis for pure cubic fields). Let $K = \mathbb{Q}(\alpha)$, where $\alpha = r^{1/3}$, for some integer $r = ab^2 \in \mathbb{Z}$ with ab square-free. Moreover, if $3 \mid r$, we assume that $3 \mid r$ and $3 \nmid b$. (These assumptions are only to ensure that ab is square-free.) Then, we have an explicit description of an integral basis for K, given as follows:

Table 5.9: Integral basis of pure cubic fields

[2] An application of Minkowski's theorem due to Hermite says that there are only finitely many number fields with a given discriminant. See [[Murthy-Esmonde], Chapter I, Section 6.5, Exercise 6.5.23]

r	$\mathcal{O}_K$
$r \not\equiv 1, 8 \mod 9$	$\mathcal{O}_K = \mathbb{Z} + \mathbb{Z}\alpha + \mathbb{Z}\frac{\alpha^2}{b}$
$r \equiv 1 \mod 9$	$\mathcal{O}_K = \mathbb{Z}\alpha + \mathbb{Z}\frac{\alpha^2}{b} + \mathbb{Z}(\frac{1+\alpha+\alpha^2}{3})$
$r \equiv 8 \mod 9$	$\mathcal{O}_K = \mathbb{Z}\alpha + \mathbb{Z}\frac{\alpha^2}{b} + \mathbb{Z}(\frac{1-\alpha+\alpha^2}{3})$

Proof. See [[Murthy-Esmonde], Chapter 4, Section 4.3, Example 4.3.6 (p. 48)]. □

We also list a method to explicitly find the integral basis of any number field. But, this method seems to be computationally hard.

THEOREM 5.3.5 (Explicit computation of an integral basis of a number field). Let K be a number field of degree n. Let $\alpha_1, \ldots, \alpha_n \in \mathcal{O}_K$ be linearly independent over $\mathbb{Q}$. Let $\Delta := d_{K/\mathbb{Q}}(\alpha_1, \ldots, \alpha_n)$. For each $1 \leq i \leq n$, we choose the least $d_{ii} \in \mathbb{N}$ so that for some integer $d_{ij} \in \mathbb{Z}$, the number $\omega_i := \frac{1}{\Delta} \cdot \sum_{j=1}^{d} d_{ij} \cdot a_j \in \mathcal{O}_K$. Then, $\omega_1, \ldots, \omega_n$ is an integral basis for K.

Proof. See [[Murthy-Esmonde], Chapter 4, Section 4.5, Exercise 4.5.17 (p. 220)]. □

We now describe an important lemma which not only helps us in proving that any number field has an integral basis, but also leads to a strategy to compute one.

LEMMA 5.3.6. Suppose that $\omega_1, \ldots, \omega_n$ is a $\mathbb{Q}-$basis for a number field K consisting of elements of $\mathcal{O}_K$. Then, $d_{K/\mathbb{Q}}(\omega_1, \ldots, \omega_n) \cdot \mathcal{O}_K \subset \mathbb{Z}\omega_1 + \ldots \mathbb{Z}\omega_n$.

Proof. See [[Jarvis], Chapter 3, Section 3.5, Lemma 3.32, (p. 54)]. □

Strategy to find an integral basis for K

Step 1: Find a $\mathbb{Q}-$basis $\{\omega_1, \ldots, \omega_n\}$ for K such that they lie in $\mathcal{O}_K$.

Step 2: Compute $\Delta := d_{K/\mathbb{Q}}(\omega_1, \ldots, \omega_n)$.

Using Lemma 5.3.6, we then have that $\mathcal{O}_K \subset \frac{1}{\Delta} \cdot (\mathbb{Z}\omega_1 + \ldots + \mathbb{Z}\omega_n)$ Thus, every algebraic integer $\alpha \in \mathcal{O}_K$ is of the form $\alpha = x_1\omega_1 + \ldots + x_n\omega_n$, where $x_i \in \mathbb{Q}$ are rational numbers whose denominators divide $\Delta, \forall\, 1 \leq i \leq n$.

Step 3: For a prime number p such that $p^2 \mid \Delta$, we check if there exists an algebraic integer $\omega' \in \mathcal{O}_K$ of the form $\omega' = x_1\omega_1 + \ldots + x_n\omega_n$ for some rational numbers x_i with denominators dividing $p, \forall\, 1 \leq i \leq n$. Now, if such an ω' exists, where some $x_i \in \mathbb{Q} \setminus \mathbb{Z}$ has denominator

p, we replace the corresponding ω_i with ω' to get a set $\{\omega_1, \ldots, \omega_{i-1}, \omega', \omega_{i+1}, \omega_n\}$, whose discriminant is now $\frac{\Delta}{p}^2$.

We now repeat **Step 3** for all primes p such that $p^2 \mid \Delta$ and then return to **Step 2**. Now, if no such ω' exists, we are left with an integral basis. For examples using this strategy, we refer to [[Jarvis], Chapter 3, Section 3.6 (p. 57)]. △

Remark 5.3.7. As a result of the above strategy, we get Proposition 5.3.1 as a Corollary.

5.3.2 Some examples

1: $K = \mathbf{Split}(X^3 - X^2 - 30X + 64; \mathbb{Q})$

Consider the splitting field K of the monic irreducible polynomial $f(X) := X^3 - X^2 - 30X + 64$ over $\mathbb{Q}$. We now normalize this cubic equation. We see that the normalized cubic is $Y^3 + pY + q$, where the numbers p and q from Section 5.3.1 are as follows:

$p = \frac{1}{3}(3b - a^2) = -\frac{91}{3}$ and $q = \frac{1}{27}(2a^3 - 9ab + 27c) = 1456/3$.

Let α be a root of $X^3 - X^2 - 30X + 64$ and $\alpha' = \alpha + \frac{1}{3}$ is the root of $Y^3 + pY + q$. We note that $\Delta = d_{K/\mathbb{Q}}(1, \alpha, \alpha^2) = d_{K/\mathbb{Q}}(1, \alpha', (\alpha')^2) = -4p^3 - 27q^2 = 2^2 \cdot 7^2 \cdot 13^2$.

We first compute the trace $T_{K/\mathbb{Q}}$ of a general element $c = x + y\alpha + z\alpha^2 \in K$, where $x, y, z \in \mathbb{Q}$. Let β and γ be the conjugates of α. We note the following:

$\alpha + \beta + \gamma = 1$

$\alpha\beta + \beta\gamma + \gamma\alpha = -30$

$\alpha\beta\gamma = -64$

Thus, as $T_{K/\mathbb{Q}}(x) = \sum_{\sigma \in \mathrm{Gal}(K/\mathbb{Q})} \sigma(x)$, for all $x \in K$, we find that

$T_{K/\mathbb{Q}}(c) = 3x + y(\alpha + \beta + \gamma) + z(\alpha^2 + \beta_3^2 + \gamma^2) = 3x + y + 61z$.

Thus, $T_{K/\mathbb{Q}}(x + y\alpha' + z(\alpha')^2) = 3x + \frac{4y}{3} + \frac{1630z}{27}$

We now claim the following: $\{1, \alpha', (\alpha')^2\}$ *is an integral basis for* K.

As $2, 7$ and 13 are the three prime factors of Δ whose squares also divide Δ, using the strategy discussed in Section 5.3.1, we are led to consider the three following cases.

Case I:

Suppose that there exists $\omega' \in \mathcal{O}_K$ of the form $\omega' = \frac{1}{2}(x_1 + x_2\alpha' + x_3(\alpha')^2)$, with $x_i \in \{0, 1\}$, for all $1 \leq i \leq 3$.

We observe that $T_{K/\mathbb{Q}}(\omega') = \frac{3x_1}{2} + \frac{2x_2}{3} + \frac{815x_3}{27}$

If $\omega' \in \mathcal{O}_K$, then $T_{K/\mathbb{Q}}(\omega') \in \mathbb{Z}$. This implies that $2 \mid x_1 \Rightarrow x_1 = 0$. Similarly, $x_2 = 0$

and $x_3 = 0$.

Case II:

Suppose that there exists $\omega' \in \mathcal{O}_K$ of the form $\omega' = \frac{1}{7}(x_1 + x_2\alpha' + x_3(\alpha')^2)$, with $x_i \in \{0, 1, \ldots, 6\}$, for all $1 \leq i \leq 3$.

We observe that $T_{K/\mathbb{Q}}(\omega') = \frac{3x_1}{7} + \frac{4x_2}{3\cdot 7} + \frac{1630x_3}{3^3\cdot 7}$. If $\omega' \in \mathcal{O}_K$, then $T_{K/\mathbb{Q}}(\omega') \in \mathbb{Z}$.

We again conclude that $x_1 = x_2 = x_3 = 0$.

Case III:

Suppose that there exists $\omega' \in \mathcal{O}_K$ of the form $\omega' = \frac{1}{13}(x_1 + x_2\alpha' + x_3(\alpha')^2)$, with $x_i \in \{0, 1, \ldots, 12\}$, for all $1 \leq i \leq 3$. Using similar techniques employed in **Case I** and **Case II**, we conclude that $x_1 = x_2 = x_3 = 0$.

Hence, $\{1, \alpha', (\alpha')^2\}$ is an integral basis for K. △

Remark 5.3.8. There exists number fields whose number rings are not free as $\mathbb{Z}-$modules on a single algebraic integer. i.e., there are number fields K such that $\mathcal{O}_K \neq \mathbb{Z}[\gamma]$, for any $\gamma \in \mathcal{O}_K$. For instance, see the discussion of the number field $K = \mathbb{Q}(\sqrt[3]{175})$ in [[Jarvis], Chapter 3, Section 3.6.4 (p. 61)].

Remark 5.3.9. We know that number field K has an integral basis; i.e., if K is a number field and $[K : \mathbb{Q}] = n$, then $\exists\, \omega_1, \ldots, \omega_n \in \mathcal{O}_K$ such that $\mathcal{O}_K = \mathbb{Z}\omega_1 \oplus \cdots \oplus \mathbb{Z}\omega_n$ [See [Murthy-Esmonde], Chapter 4, Section 4.2, Exercise 4.2.3 (p. 45)]. We can extend this result to a finite extension of number fields L/K whenever $\mathcal{O}_K$ is a PID and prove similarly just like the case of $K/\mathbb{Q}$ that $\mathcal{O}_L$ is free as a $\mathcal{O}_K-$module. However, if we have a finite extension L/K of number fields such that $\mathcal{O}_K$ is not a PID, then $\mathcal{O}_L$ need not be free as a $\mathcal{O}_K$-module. See [Conrad-NFRIE] for more details.

5.4 Some examples of degree 3 abelian extensions of $\mathbb{Q}$

5.4.1 Example 1: $\mathbb{Q}(\zeta_7 + \zeta_7^6)/\mathbb{Q}$

Let $\zeta := \zeta_7 = e^{\frac{2\pi i}{7}}$ denote the primitive 7th-root of unity.

We claim the following:

The extension $\mathbb{Q}(\zeta + \zeta^6)/\mathbb{Q}$ is an abelian extension of degree 3 over $\mathbb{Q}$.

We first observe that as $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$, $\mathbb{Q}(\zeta)/\mathbb{Q}$ is an abelian extension of degree 6, whose Galois group $G \cong \mathbb{Z}/6/Z$. Using basic Galois theory, we next find a normal

subgroup of index 3 (order 2) in G and consider its fixed field which would be our required degree 3 abelian extension.

We note the following: $\langle \bar{1}, \bar{6} \rangle \leqslant (\mathbb{Z}/7\mathbb{Z})^\times$ is a subgroup of order 2. The corresponding subgroup H of $\mathrm{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q})$ of order 2 is the one generated by complex-conjugation σ. i.e., $H = \langle \sigma \rangle$, where,

$$\begin{aligned} \sigma \colon \mathbb{Q}(\zeta_7) &\to \mathbb{Q}(\zeta_7) \\ \zeta_7 &\mapsto \zeta_7^6 = \zeta_7^{-1} \\ (1 &\mapsto 1) \end{aligned}$$

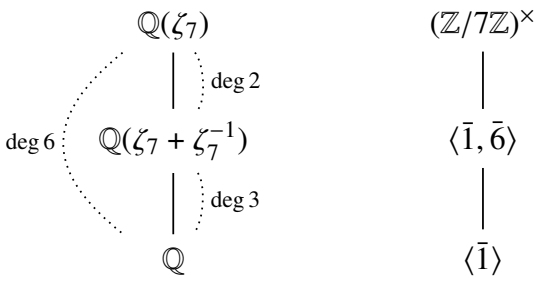
We see that $\alpha := \zeta_7 + \zeta_7^6$ is fixed by H. We calculate that

$$\begin{aligned} \alpha^3 &= \zeta_7^3 + \zeta_7^{-3} + 3\zeta_7 + 3\zeta_7^{-1} \\ \alpha^2 &= \zeta_7^2 + \zeta_7^{-2} + 2 \end{aligned}$$

Hence, α satisfies the equation $X^3 + X^2 - 2X - 1 = 0$. We also note using the rational root test that the polynomial $f(X) := X^3 + X^2 - 2X - 1 \in \mathbb{Z}[X]$ is irreducible over $\mathbb{Q}$.

Hence, the sub-extension $\mathbb{Q}(\zeta_7)^H = \mathbb{Q}(\alpha)$ is a degree 3 abelian extension of $\mathbb{Q}$. The conjugates of $\alpha := \zeta_7 + \zeta_7^6$ are $\alpha^{(1)} = \alpha = \zeta_7 + \zeta_7^6$, $\alpha^{(2)} = \zeta_7^2 + \zeta_7^5 = \alpha^2$ and $\alpha^{(3)} = \zeta_7^3 + \zeta_7^4 = \alpha^3$.

We have the following tower and lattice diagram.



We now use Proposition 3.0.2 and we also calculate the discriminant.

As 7 is the only ramified rational prime in $\mathbb{Q}(\zeta_7)$, the discriminant $d_{K/\mathbb{Q}}$ is a power of 7. Using Proposition 3.0.4 we get that the discriminant

$$d_{\mathbb{Q}(\zeta_7+\zeta_7^{-1})/\mathbb{Q}} = N_{\mathbb{Q}(\zeta_7+\zeta_7^{-1})/\mathbb{Q}}(3\alpha^2 + 2\alpha - 2) = 7^2$$

We now calculate the Minkowski bound M and use the same to calculate the ideal class group of K. We use the formula $M = \frac{n!}{n^n} \cdot \left(\frac{4}{\pi}\right)^{r_2} \cdot \sqrt{|d_K|}$ Here, $n = 3, r_2 = 0, d_K = 7^2$. Therefore, $M \approx 1.56$. As the Minkowski bound is less than 2, the class group is trivial. □

5.4.2 Example 2: $\mathbb{Q}(\zeta_{13} + \zeta_{13}^5 + \zeta_{13}^{12} + \zeta_{13}^8)/\mathbb{Q}$

Let $\zeta := \zeta_{13} = e^{\frac{2\pi i}{13}}$ denote the primitive 13th-root of unity. Let $G = \text{Gal}(\mathbb{Q}(\zeta_{13})/\mathbb{Q})$.

We claim the following: *The extension* $\mathbb{Q}(\zeta_{13}+\zeta_{13}^5+\zeta_{13}^{12}+\zeta_{13}^8)/\mathbb{Q}$ *is an abelian extension of degree* 3 *over* $\mathbb{Q}$.

We first look for an index 3 subgroup $H \leqslant G$. Hence, $|H| = 4$. We see that $\bar{5} \in (\mathbb{Z}/13\mathbb{Z})^\times$ such that the order of $\bar{5}$ is 4. $\langle\bar{5}\rangle = \{\bar{5}, \bar{12}, \bar{8}, \bar{1}\} \leqslant (\mathbb{Z}/13\mathbb{Z})^\times$. We let $H := \langle\sigma\rangle$ where $\sigma \in G$ is given by,

$$\begin{gathered}\sigma\colon \mathbb{Q}(\zeta_{13}) \to \mathbb{Q}(\zeta_{13}) \\ \zeta_{13} \mapsto \zeta_{13}^5 \\ (1 \mapsto 1)\end{gathered}$$

Thus, if $\mathbb{Q}(\zeta_{13})^H$ is our required degree 3 abelian extension contained in the cyclotomic extension $\mathbb{Q}(\zeta_{13})$.

We observe that the element $\alpha := \zeta_{13} + \zeta_{13}^5 + \zeta_{13}^{12} + \zeta_{13}^8$ is fixed by H . We also note that α satisfies the polynomial $X^3 + X^2 - 4X + 1 \in \mathbb{Z}[X]$.

We thus calculate the discriminant $(d_K) = N_{K/\mathbb{Q}}(\mathfrak{D}_{K/\mathbb{Q}}) = N_{K/\mathbb{Q}}(f'(\alpha))$, where $f(X)$ is the minimal polynomial of α over $\mathbb{Q}$.

We find that $d_K = N_{K/\mathbb{Q}}(3\alpha^2 + 2\alpha - 4) = 13^2$. We now calculate the Minkowski bound M and use the same to calculate the ideal class group of K. We use the formula $M = \frac{n!}{n^n} \cdot (\frac{4}{\pi})^{r_2} \cdot \sqrt{|d_K|}$ Here, $n = 3, r_2 = 0, d_K = 13^2$. Therefore, $M \approx 2.89$. We now calculate the ideal class group of K . The ideal class group is generated by the prime ideals with norm not exceeding the bound $M \approx 2.89$. Hence, we require the prime factorization of the ideal (2) in $\mathcal{O}_K$. We verify that $2 \nmid \Delta(f)$

We now use Dedekind's criterion (Theorem 3.1.3).

Table 5.11: Factorization of some ideals in $\mathcal{O}_K$, where $K = \text{Split}(X^3 + X^2 - 4X + 1; \mathbb{Q})$

prime number p	Irreducible factors of (X^3+X^2-4X+1) mod p	(p) in $\mathcal{O}_K$
2	$X^3 + X^2 + 1$ *Irreducible*	*prime*

As (2) remains prime in $\mathcal{O}_K$, (2) $\sim$ (1), and we get that the ideal class group of K is trivial. □

5.4.3 Example 3: Degree 3 sub-extensions of $\mathbb{Q}(\zeta_{91})$

We now analyze the degree 3 abelian extensions contained in the cyclotomic extension $\mathbb{Q}(\zeta_{91})$.

Let $K := \mathbb{Q}(\zeta_{91}) = \mathbb{Q}(\zeta_{13}, \zeta_7)$.

We first note that the Galois group $G = \mathrm{Gal}(\mathbb{Q}(\zeta_{91})/\mathbb{Q}) = (\mathbb{Z}/(7)))^\times \times (\mathbb{Z}/(13))^\times \cong \mathbb{Z}/(2) \times \mathbb{Z}/(3) \times \mathbb{Z}/(3) \times \mathbb{Z}/(4)$. Here, we use the fact that the multiplicative group of a finite field is cyclic, and that $\mathbb{Z}/(mn) \cong \mathbb{Z}/(m) \times \mathbb{Z}/(n) \iff (m, n) = 1$.

We want subgroups H such that $\mathrm{Gal}(K^H)/\mathbb{Q} \cong \mathbb{Z}/(3)$. We are first led to consider the subgroups of $\mathbb{Z}/(3) \times \mathbb{Z}/(3)$ which are of order 3. We claim that there are 4 subgroups of order 3 in $\mathbb{Z}/(3) \times \mathbb{Z}/(3)$. This is because the number of $1-$dimensional linear subspaces in $\mathbb{F}_p^2$ are $p + 1$ in number.

We denote these 4 degree 3 subspaces in $\mathbb{Z}/(3) \times \mathbb{Z}/(3)$ as $H_1 := (0) \times \mathbb{Z}/(3)$, $H_2 := \mathbb{Z}/(3) \times (0)$, $H_3 := \{(1, 2), (2, 1), (0, 0)\}$, and $H_4 := \{(1, 1), (2, 2), (0, 0)\}$. We also note the following: In $(\mathbb{Z}/(7))^\times$, the degree 3 subgroup is $\langle \bar{2}, \bar{4}, \bar{1} \rangle$, and the degree 2 subgroup is $\langle \bar{6}, \bar{1} \rangle$. In $(\mathbb{Z}/(13))^\times$, the degree 4 subgroup is $\langle \bar{5}, \bar{12}, \bar{8}, \bar{1} \rangle$, and the degree 3 subgroup is $\langle \bar{3}, \bar{9}, \bar{1} \rangle$. As $(\mathbb{Z}/(7))^\times \times (\mathbb{Z}/(13))^\times \cong \mathbb{Z}/(2) \times \mathbb{Z}/(3) \times \mathbb{Z}/(3) \times \mathbb{Z}/(4)$, we have the following correspondence:

$$\begin{array}{c} \mathbb{Z}/(2) \times \mathbb{Z}/(3) \times \mathbb{Z}/(3) \times \mathbb{Z}/(4) \\ \updownarrow \\ (\mathbb{Z}/(7))^\times \times (\mathbb{Z}/(13))^\times \end{array}$$

$$\begin{array}{ccc} (0) \times \mathbb{Z}/(3) & \mathbb{Z}/(3) \times (0) & \{(1, 2), (2, 1), (0, 0)\} \\ \updownarrow & \updownarrow & \updownarrow \\ (0) \times \{\bar{3}, \bar{9}, \bar{1}\} & \{\bar{2}, \bar{4}, \bar{1}\} \times (0) & \{(\bar{2}, \bar{3}), (\bar{4}, \bar{9}), (\bar{1}, \bar{1})\} \end{array}$$

$$\begin{array}{c} \{(1, 1), (2, 2), (0, 0)\} \\ \updownarrow \\ \{(\bar{2}, \bar{9}), (\bar{4}, \bar{3}), (\bar{1}, \bar{1})\} \end{array}$$

We now describe the four degree-3 abelian extensions contained in K.

Let G_i denote the index 3 subgroups of G such that $G/G_i \cong H_i$, for $i = 1, 2, 3, 4$.

Since $\mathrm{Gal}(\mathbb{Q}(\zeta_{91})/\mathbb{Q}) \cong \mathrm{Gal}(\mathbb{Q}(\zeta_{13})/\mathbb{Q}) \times \mathrm{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q})$, we adopt the convention to

denote by σ or its variants an element of G which restricts to an element of $\mathrm{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q})$, and by τ or its variants an element of G which restricts to an element of $\mathrm{Gal}(\mathbb{Q}(\zeta_{13})/\mathbb{Q})$.

We define the following automorphisms:

Complex conjugation $\sigma_{\bar{6}}$ of order 2

$$\begin{aligned}\sigma_{\bar{6}}\colon \mathbb{Q}(\zeta_{13},\zeta_7) &\to \mathbb{Q}(\zeta_{13},\zeta_7)\\ \zeta_{13} &\mapsto \zeta_{13}\\ \zeta_7 &\mapsto \zeta_7^6 = \zeta_7^{-1}\\ (1 &\mapsto 1)\end{aligned}$$

Automorphism $\sigma_{\bar{2}}$ of order 3

$$\begin{aligned}\sigma_{\bar{2}}\colon \mathbb{Q}(\zeta_{13},\zeta_7) &\to \mathbb{Q}(\zeta_{13},\zeta_7)\\ \zeta_{13} &\mapsto \zeta_{13}\\ \zeta_7 &\mapsto \zeta_7^2\\ (1 &\mapsto 1)\end{aligned}$$

Automorphism $\tau_{\bar{5}}$ of order 4

$$\begin{aligned}\tau_{\bar{5}}\colon \mathbb{Q}(\zeta_{13},\zeta_7) &\to \mathbb{Q}(\zeta_{13},\zeta_7)\\ \zeta_{13} &\mapsto \zeta_{13}^5\\ \zeta_7 &\mapsto \zeta_7\\ (1 &\mapsto 1)\end{aligned}$$

Automorphism $\tau_{\bar{3}}$ of order 3

$$\begin{aligned}\tau_{\bar{3}}\colon \mathbb{Q}(\zeta_{13},\zeta_7) &\to \mathbb{Q}(\zeta_{13},\zeta_7)\\ \zeta_{13} &\mapsto \zeta_{13}^3\\ \zeta_7 &\mapsto \zeta_7\\ (1 &\mapsto 1)\end{aligned}$$

Sub-extension 1

$G_1 := \mathbb{Z}/(2) \times \mathbb{Z}/(3) \times (0) \times \mathbb{Z}/(4) \leqslant G$.

We note that G_1 has 24 elements and hence index 3 in G.

$$G/G_1 \cong (0) \times \mathbb{Z}/(3) = H_1$$

We note that $G_1 = \langle \sigma_{\bar{6}}, \sigma_{\bar{2}}, \tau_{\bar{5}} \rangle$.

The fixed field $K_1 := \mathbb{Q}(\zeta_{91})^{G_1}$ of G_1 in $\mathbb{Q}(\zeta_{91})$ is the required degree 3 abelian sub-extension.

Since G_1 contains both $\sigma_{\bar{6}}$ and $\sigma_{\bar{2}}$, the maximal sub-field of $\mathbb{Q}(\zeta_7)$ in $\mathbb{Q}(\zeta_{91})$ fixed by G_1 is $\mathbb{Q}$.

Now, the element $\alpha_1 := \zeta_{13} + \zeta_{13}^5 + \zeta_{13}^{12} + \zeta_{13}^8$ is fixed by $\tau_{\bar{5}}$ and hence also by G_1 . We also note that α_1 satisfies the polynomial $X^3 + X^2 - 4X + 1$, which is irreducible over $\mathbb{Q}$.

As the index of G_1 in G is 3, we conclude that the fixed field $K_1 := \mathbb{Q}(\zeta_{91})^{G_1}$ of G_1 in $\mathbb{Q}(\zeta_{91})$ is $K_1 = \mathbb{Q}(\alpha_1)$, due to the calculations from Section 5.4.2. The Minkowski bound M was calculated to be approximately 2.89 and was shown that the ideal class group of K_1 is trivial.

Sub-extension 2

$G_2 := \mathbb{Z}/(2) \times (0) \times \mathbb{Z}/(3) \times \mathbb{Z}/(4) \leqslant G$.

We note that G_2 = has 24 elements and hence index 3 in G.

$$G/G_2 \cong \mathbb{Z}/(3) \times (0) = H_2$$

We note that $G_2 = \langle \sigma_{\bar{6}}, \tau_{\bar{5}}, \tau_{\bar{3}} \rangle$. The fixed field $K_2 := \mathbb{Q}(\zeta_{91})^{G_2}$ of G_2 in $\mathbb{Q}(\zeta_{91})$ is the required degree 3 abelian sub-extension. Since G_2 contains both $\tau_{\bar{5}}$ and $\tau_{\bar{3}}$, the maximal sub-field of $\mathbb{Q}(\zeta_{13})$ in $\mathbb{Q}(\zeta_{91})$ fixed by G_2 is $\mathbb{Q}$. Now, the element $\alpha_2 := \zeta_7 + \zeta_7^6$ is fixed by $\sigma_{\bar{6}}$ and hence also by G_2 . We also note that α_2 satisfies the polynomial $X^3 + X^2 - 2X - 1$, which is irreducible over $\mathbb{Q}$. Hence, the sub-extension $\mathbb{Q}(\zeta_{91})^{G_2} = \mathbb{Q}(\alpha_2)$ is a degree 3 abelian extension of $\mathbb{Q}$. The conjugates of $\alpha := \zeta_7 + \zeta_7^6$ are $\alpha^{(1)} = \alpha = \zeta_7 + \zeta_7^6$, $\alpha^{(2)} = \zeta_7^2 + \zeta_7^5 = \alpha^2$ and $\alpha^{(3)} = \zeta_7^3 + \zeta_7^4 = \alpha^3$. As the index of G_2 in G is 3, we conclude that the fixed field $K_2 := \mathbb{Q}(\zeta_{91})^{G_2}$ of G_2 in $\mathbb{Q}(\zeta_{91})$ is $K_2 = \mathbb{Q}(\alpha_2)$, due to the calculations from Section 5.4.1. The Minkowski bound M was calculated to be approximately 1.56 and was shown that the ideal class group of K_2 is trivial.

Sub-extension 3

We wish to find a subgroup G_3 of index 3 in G such that $G_3 \cong \mathbb{Z}/(2) \times H_3 \times \mathbb{Z}/(4)$, where $H_3 = \{(1,2),(2,1),(0,0)\} \leqslant \mathbb{Z}/(3) \times \mathbb{Z}/(3)$.

We have the following correspondence as $(\mathbb{Z}/(7))^\times \times (\mathbb{Z}/(13))^\times \cong \mathbb{Z}/(2) \times \mathbb{Z}/(3) \times \mathbb{Z}/(3) \times \mathbb{Z}/(4)$

$$\{(1,2),(2,1),(0,0)\} \leqslant \mathbb{Z}/(3) \times \mathbb{Z}/(3)$$
$$\updownarrow$$
$$\{(\bar{2},\bar{3}),(\bar{4},\bar{9}),(\bar{1},\bar{1})\} \leqslant (\mathbb{Z}/(7))^\times \times (\mathbb{Z}/(13))^\times$$

Hence, we define the following automorphism:

Automorphism $\gamma_{(\bar{2},\bar{3})}$ of order 3:

$$\begin{aligned}\gamma_{(\bar{2},\bar{3})}\colon \mathbb{Q}(\zeta_{13},\zeta_7) &\to \mathbb{Q}(\zeta_{13},\zeta_7)\\ \zeta_7 &\mapsto \zeta_7^2\\ \zeta_{13} &\mapsto \zeta_{13}^3\\ (1 &\mapsto 1)\end{aligned}$$

We now define $G_3 \leqslant G$ in the following manner:

$$G_3 := \langle\, \sigma_{\bar{6}}, \gamma_{(\bar{2},\bar{3})}, \tau_{\bar{5}} \,\rangle,$$

where, $\sigma_{\bar{6}}$ restricts to the complex conjugation automorphism of order 2 for $\mathbb{Q}(\zeta_7)$ and $\tau_{\bar{5}}$ restricts to the automorphism of order 4 for $\mathbb{Q}(\zeta_{13})$. G_3 is of order 24 and hence, index 3 in G.

We wish to find the fixed field $K_3 := \mathbb{Q}(\zeta_{91})^{G_3}$ which is a degree 3 abelian extension.

We achieve the same in 3 steps. We find the fixed field of the next generator in subsequent steps.

We let $a := \zeta_7$ and $b := \zeta_{13}$. Computations are done using **SageMath**. We refer to [Stein] and [Sage] for the same.

Remark 5.4.1. We are not using any information from **Sage** which is unknown to us. Finding the minimal polynomial of an algebraic element θ over $\mathbb{Q}$ can be done using the *regular representation* of $\mathbb{Q}(\theta)$ over $\mathbb{Q}$ with respect to a $\mathbb{Q}-$basis of $\mathbb{Q}(\theta)$. The minimal polynomial of θ would then be the characteristic polynomial of the matrix representing θ. We refer to [[Janusz], Chapter 1, Section 7, *'Factorization in a Cubic Extension'*, p. 41] for details and a worked out example.

Step 1:

We observe that the element $\beta := a^2b^3 + a^4b^9 + ab$ is fixed by $\langle\gamma_{(\bar{2},\bar{3})}\rangle$. We find the minimal polynomial of β over $\mathbb{Q}$ using the following code.

```
K.<b> = CyclotomicField(13)
L.<a> = K.extension(1+x+x^2+x^3+x^4+x^5+x^6)
beta=a^2*b^3+a^4*b^9+a*b
beta.absolute_minpoly('t')
```

We find from above that the minimal polynomial $f_1(t)$ of β over $\mathbb{Q}$ is

$$\begin{aligned} f_1(t) = t^{24} - t^{23} - t^{22} - 49t^{21} + 51t^{20} + 60t^{19} + 800t^{18} - 855t^{17} - 1148t^{16} - 5098t^{15} \\ + 5795t^{14} + 8184t^{13} + 12089t^{12} - 16093t^{11} - 16028t^{10} - 17358t^9 + 4096t^8 \\ + 16593t^7 + 30608t^6 + 13582t^5 - 2206t^4 - 24087t^3 + 3890t^2 + 7650t + 2549. \end{aligned}$$

Now, as $\langle \gamma_{(\bar{2},\bar{3})} \rangle$ is of index 24 in G, we conclude that $\mathbb{Q}(\zeta_{91})^{\langle \gamma_{(\bar{2},\bar{3})} \rangle} = \mathbb{Q}(\beta) = \mathbb{Q}(a^2b^3 + a^4b^9 + ab)$.

Remark 5.4.2. One might also observe that the element $a^2+b^3+a^4+b^9+a+b$ is fixed by $\gamma_{(\bar{2},\bar{3})}$. But, we note that $a^2 + b^3 + a^4 + b^9 + a + b$ is also fixed by $\tau_{\bar{3}} \notin G_3$, which we do not want. In fact, using **Sage** , we find that the minimal polynomial of $a^2 + b^3 + a^4 + b^9 + a + b$ over $\mathbb{Q}$ is of degree 8.

Step 2:

We now find a sub-field of $\mathbb{Q}(a^2b^3 + a^4b^9 + ab)$ which is fixed by one of the other generators of G_3, say, $\sigma_{\bar{6}}$.

We note that the conjugate of $a^2b^3 + a^4b^9 + ab$ under $\sigma_{\bar{6}}$ is $a^5b^3 + a^3b^9 + a^6b$.

We now find the minimal polynomial of the element $\beta_2 := a^2b^3 + a^4b^9 + ab + a^5b^3 + a^3b^9 + a^6b$ on **Sage** using the following code.

```
K.<b> = CyclotomicField(13)
L.<a> = K.extension(1+x+x^2+x^3+x^4+x^5+x^6)
beta2=a^2*b^3+a^4*b^9+a*b + a^5*b^3 + a^3*b^9 + a^6*b
beta2.absolute_minpoly('t')
```

We find that the minimal polynomial $f_2(t)$ of β_2 over $\mathbb{Q}$ is

$$\begin{aligned} f_2(t) = t^{12} - t^{11} + t^{10} - 27t^9 + 27t^8 + 90t^7 + 53t^6 - 1353t^5 \\ + 768t^4 + 3886t^3 + 1600t^2 - 5409t + 1847. \end{aligned}$$

Now, we note that as $\langle \gamma_{(\bar{2},\bar{3})}, \sigma_{\bar{6}} \rangle$ is of index 12 in G, we conclude that $\mathbb{Q}(\zeta_{91})^{\langle \gamma_{(\bar{2},\bar{3})}, \sigma_{\bar{6}} \rangle} = \mathbb{Q}(\beta_2) = \mathbb{Q}(a^2b^3 + a^4b^9 + ab + a^5b^3 + a^3b^9 + a^6b)$.

Step 3:

We finally find the sub field of $\mathbb{Q}(\beta_2)$ which is fixed by the generator $\tau_{\bar{5}}$ of G_3.

We first find the conjugates of β_2 under $\tau_{\bar{5}}$.

The conjugates of β_2 under $\tau_{\bar{5}}$ are:

$a^2b^3 + a^4b^9 + ab + a^5b^3 + a^3b^9 + a^6b$, $a^2b^2 + a^4b^6 + ab^5 + a^5b^2 + a^3b^6 + a^6b^5$, $a^2b^{10} + a^4b^4 + ab^{12} + a^5b^{10} + a^3b^4 + a^6b^{12}$, and $a^2b^{11} + a^4b^7 + ab^8 + a^5b^{11} + a^3b^7 + a^6b^8$.

Let $\beta_3 := \beta_2 + \tau_{\bar{5}}(\beta_2) + \tau_{\bar{5}}^2(\beta_2) + \tau_{\bar{5}}^3(\beta_2)$

i.e., $\beta_3 := a^2b^3 + a^4b^9 + ab + a^5b^3 + a^3b^9 + a^6b + a^2b^2 + a^4b^6 + ab^5 + a^5b^2 + a^3b^6 + a^6b^5 + a^2b^{10} + a^4b^4 + ab^{12} + a^5b^{10} + a^3b^4 + a^6b^{12} + a^2b^{11} + a^4b^7 + ab^8 + a^5b^{11} + a^3b^7 + a^6b^8$.

We find the minimal polynomial $f_3(X)$ of β_3 on **Sage** and find that it is of degree 3.

```
K.<b> = CyclotomicField(13)
L.<a> = K.extension(1+x+x^2+x^3+x^4+x^5+x^6)
beta3= a^2*b^3+a^4*b^9+a*b + a^5*b^3+a^3*b^9+a^6*b + a^2*b^2 +a^4*b^6 +
    a*b^5 + a^5*b^2 + a^3*b^6+a^6*b^5 + a^2*b^10 + a^4*b^4 + a*b^12 + a
    ^5*b^10 + a^3*b^4 + a^6*b^12 + a^2*b^11 + a^4*b^7 + a*b^8 + a^5*b^11
     + a^3*b^7 + a^6*b^8
beta3.absolute_minpoly('X')
```

The minimal polynomial $f_3(X)$ of β_3 over $\mathbb{Q}$ is found using the above code on **Sage** to be $f_3(X) = X^3 - X^2 - 30X + 64$.

Hence, we conclude that $K_3 := \mathbb{Q}(\zeta_{91})^{G_3} = \mathbb{Q}(\beta_3)$.

We now compute the Minkowski bound and the ideal class group of K_3.

We first calculate the discriminant.

We note that as $K_3 \subset \mathbb{Q}(\zeta_{91}) = \mathbb{Q}(\zeta_{13}, \zeta_7)$, and the only ramified primes in $\mathbb{Q}(\zeta_{91})$ are 7 and 13, the discriminant d_{K_3} of K_3 must have only 7 and 13 as its prime factors.

We have the following tower diagram:

$$
\begin{array}{c}
\mathbb{Q}(\zeta_{91}) \\
\vdots\ \deg 3 \\
\mathbb{Q}(a^2b^3 + a^4b^9 + ab) = \mathbb{Q}(\zeta_{91})^{\langle \gamma_{(\bar{2},\bar{3})} \rangle} \\
\vdots\ \deg 2 \\
\mathbb{Q}(\beta_2) = \mathbb{Q}(\zeta_{91})^{\langle \gamma_{(\bar{2},\bar{3})}, \sigma_{\bar{6}} \rangle} \\
\vdots\ \deg 4 \\
\mathbb{Q}(\beta_3) = \mathbb{Q}(\zeta_{91})^{\langle\ \gamma_{(\bar{2},\ \bar{3})},\ \sigma_{\bar{6}},\ \tau_{\bar{5}}\ \rangle} \\
\vdots\ \deg 3 \\
\mathbb{Q}
\end{array}
$$

By using Proposition 3.0.3 we get that,

$$d_{\mathbb{Q}(\zeta_{91})/\mathbb{Q}} = (d_{\mathbb{Q}(\beta_3)/\mathbb{Q}})^{24} \cdot N_{\mathbb{Q}(\beta_3)/\mathbb{Q}}\,(d_{\mathbb{Q}(\zeta_{91})/\mathbb{Q}(\beta_3)}) \tag{5.2}$$

We note from Minkowski's theorem that at least one of 7 and 13 are ramified in K_3, and these are the only ramified primes in K_3. (*) (we know that 7 and 13 are the only primes that should be ramified in $\mathbb{Q}(\beta_3)$ by analyzing $d_{\mathbb{Q}(\zeta_{91})}$, whose only prime factors are 7 and 13.)

Let $m := [O_{K_3} : \mathbb{Z}[\beta_3]] = |O_{K_3}/\mathbb{Z}[\beta_3]|$. We use the formula,

$$d_{\mathbb{Q}(\beta_3)/\mathbb{Q}}(\beta_3) = m^2 \cdot d_{\mathbb{Q}(\beta_3)/\mathbb{Q}} \tag{5.3}$$

We find that the conjugates of β_3 are $\frac{\beta_3^2}{2} + \frac{\beta_3}{2} - 10$ and $-\frac{\beta_3^2}{2} - \frac{3\beta_3}{2} + 11$. We calculate $d_{\mathbb{Q}(\zeta_{91})}(\beta_3) = 2^2 \cdot 7^2 \cdot 13^2$ using the formula $d_{\mathbb{Q}(\zeta_{91})}(\beta_3) = (-1)^{\binom{n}{2}} \cdot \prod_{i=1}^{n} f'(\beta_3^{(i)})$ (here, $n = [\mathbb{Q}(\beta_3) : \mathbb{Q}] = 3$ and $f(X)$ is the minimal polynomial of β_3 over $\mathbb{Q}$.) We note by comparing the coefficients in Equation (5.3) that odd powers of 7 and 13 cannot divide d_K. By noting that $d_{\mathbb{Q}(\zeta_{91})/\mathbb{Q}} = 7^{60} \cdot 13^{66}$, the statement (*) , and comparing the coefficients of 7 and 13 in Equation (5.2), we are left with the following choices for d_{K_3}: 7^2, or 13^2, or $7^2 \cdot 13^2$. Hence, we conclude that, $d_{\mathbb{Q}(\beta_3)/\mathbb{Q}} \leq 7^2 \cdot 13^2$.

We now calculate the Minkowski bound M for K_3:

We use the formula $M = \frac{n!}{n^n} \cdot (\frac{4}{\pi})^{r_2} \cdot \sqrt{|d_{K_3}|}$ Here, $n = 3, r_2 = 0, d_{K_3} \leq 7^2 \cdot 13^2$. We find that $M \lessapprox 20.22$.

We note that 2, 3, 5, 7, 11, 13, 17, and 19 are all the rational primes less than or equal to M.

We now calculate the ideal class group of K_3 . The ideal class group is generated by the prime ideals with norm not exceeding the bound $M \approx 20.22$. Hence, we require the prime factorization of the ideals (3), (5), (11), (17), and (19) in O_{K_3}, for which we use Dedekind's criterion (Theorem 3.1.3). Note that we cannot use Dedekind's criterion for the ideals (2), (7), and (13), as they contain $d_{\mathbb{Q}(\zeta_{91})}(\beta_3)$. We use the algorithm in Proposition 3.1.2 to compute the prime factorization of the ideals (2), (7), and (13).

We now have the following: We note that $N_{K_3/\mathbb{Q}}(\beta_3) = -64$, and

$N_{K_3/\mathbb{Q}}(\beta_3 + 1) = \det([m_{\beta_3+1}]) = \det\left(\begin{bmatrix} 1 & 0 & -64 \\ 1 & 1 & 30 \\ 0 & 1 & 2 \end{bmatrix}\right) = -92$, where m_{β_3+1} is the $\mathbb{Q}-$linear multiplication by β_3 map.

Since $2 \mid N_{K_3/\mathbb{Q}}(\beta_3)$ and $2 \mid N_{K_3/\mathbb{Q}}(\beta_3 + 1)$, using Proposition 3.1.2, we conclude that 2 has at least two distinct primes $\mathfrak{p}_2$ and $\mathfrak{p}_2'$ of residue field degree 1 lying over it.

Since $e_2 \cdot f_2 \cdot g_2 = 3$ and as 2 is unramified in K_3, we conclude that $f_2 = 1$, and hence

that 2 has exactly three primes which lie over it and it is completely split in K_3. i.e., $(2) = \mathfrak{p}_2 \cdot \mathfrak{p}_2' \cdot \mathfrak{p}_2''$.

Below is the **Sage** code to compute the norm of the element $\beta_3 + i$, for some integer $i \in \mathbb{Z}$:

```
K.<beta_3> = NumberField(x^3-x^2-30*x+64)
I = K.fractional_ideal(beta_3+i)
factor(I.norm())
```

Similarly, we compute the following for $p = 7$:

$$N_{K_3/\mathbb{Q}}(\beta_3) = -2^6 \qquad N_{K_3/\mathbb{Q}}(\beta_3 + 1) = -2^2 \cdot 23 \qquad N_{K_3/\mathbb{Q}}(\beta_3 + 2) = 7 \cdot 2^4$$

$$N_{K_3/\mathbb{Q}}(\beta_3 + 3) = 2 \cdot 59 \qquad N_{K_3/\mathbb{Q}}(\beta_3 + 4) = 2^3 \cdot 13 \qquad N_{K_3/\mathbb{Q}}(\beta_3 + 5) = 2^6$$

$$N_{K_3/\mathbb{Q}}(\beta_3 + 6) = 2^3$$

We note that 7 only divides $N_{K_3/\mathbb{Q}}(\beta_3 + 2)$ among $N_{K_3/\mathbb{Q}}(\beta_3 + i)$, for $0 \leq i \leq 6$. Hence, using Proposition 3.1.2, we conclude that there is a unique prime $\mathfrak{p}_7 \subset \mathcal{O}_{K_3}$ which lies over 7. We further know that 7 is ramified in K_3. Hence, as $e_7 \cdot f_7 \cdot g_7 = 3$ and $g_7 = 1$ and $e_7 > 1$, we conclude that $f_7 = 1, e_7 = 3$, and hence, $(7) = \mathfrak{p}_7^3$.

We finally compute the norm to find the prime factorization of (7). We make the following additional computations:

$$N_{K_3/\mathbb{Q}}(\beta_3 + 7) = 2 \cdot 59 \qquad N_{K_3/\mathbb{Q}}(\beta_3 + 8) = 2^4 \cdot 17 \qquad N_{K_3/\mathbb{Q}}(\beta_3 + 9) = 2^2 \cdot 7 \cdot 17$$

$$N_{K_3/\mathbb{Q}}(\beta_3 + 10) = 2^5 \cdot 23 \qquad N_{K_3/\mathbb{Q}}(\beta_3 + 11) = 2 \cdot 23^2 \qquad N_{K_3/\mathbb{Q}}(\beta_3 + 12) = 2^3 \cdot 181$$

We similarly note that 13 only divides $N_{K_3/\mathbb{Q}}(\beta_3+4)$ among $N_{K_3/\mathbb{Q}}(\beta_3+i)$, for $0 \leq i \leq 12$. Hence, using Proposition 3.1.2, we conclude that there is a unique prime $\mathfrak{p}_{13} \subset \mathcal{O}_{K_3}$ which lies over 13. We further know that 13 is ramified in K_3. Hence, as $e_{13} \cdot f_{13} \cdot g_{13} = 3$ and $g_{13} = 1$ and $e_{13} > 1$, we conclude that $(13) = \mathfrak{p}_{13}^3$.

We thus have found that $(2) = \mathfrak{p}_2 \cdot \mathfrak{p}_2' \cdot \mathfrak{p}_2''$, $(7) = \mathfrak{p}_7^3$, and $(13) = \mathfrak{p}_{13}^3$.

We now proceed to find the prime factorization of the ideals (3), (5), (11), (17), and (19) in $\mathcal{O}_{K_3}$ using Dedekind's criterion (Theorem 3.1.3). Below is the **Sage** code to find the factorization of a polynomial modulo a prime, for instance, 17:

```
X = PolynomialRing(GF(17), 'X').gen()
f = X^3 - X^2 - 30*X + 64
f.factor()
```

We finally have the following table:

Table 5.15: Factorization of some ideals in O_{K_3}, where $K_3 = \text{Split}(X^3 - X^2 - 30X + 64; \mathbb{Q})$

prime number p	Irreducible factors of $(X^3 - X^2 - 30X + 64) \mod p$	Factorization of (p) in O_{K_3}
2	$(X+1) \cdot X^2$	$\mathfrak{p}_2 \cdot (\mathfrak{p}'_2)^2$: wrong conclusion as we cannot use Dedekind's criterion in this case. The correct factorization is $\mathfrak{p}_2 \cdot \mathfrak{p}'_2 \cdot \mathfrak{p}''_2$
3	$X^3 + 2X^2 + 1$ *(Irreducible)*	*prime*
5	$X^3 + 4X^2 + 4$ *(Irreducible)*	*prime*
7	$(X+2)^3$	$\mathfrak{p}_7^3$: the factorization is correct, but Dedekind's criterion can't be used here
11	$X^3 + 10X^2 + 3X + 9$ *(Irreducible)*	*prime*
13	$(X+4)^3$	$\mathfrak{p}_{13}^3$: the factorization is correct, but Dedekind's criterion can't be used here
17	$(X+8)(X+9)(X+16)$	$\mathfrak{p}_{17} \cdot \mathfrak{p}'_{17} \cdot \mathfrak{p}''_{17}$
19	$X^3 + 18X^2 + 8X + 7$ *(Irreducible)*	*prime*

We thus note that (3) ~ (5) ~ (11) ~ (19) ~ (1) and hence, the ideal class group $\text{Cl}(K_3)$ of K_3 is generated by the ideal classes $[\mathfrak{p}_2], [\mathfrak{p}'_2], [\mathfrak{p}''_2], [\mathfrak{p}_7], [\mathfrak{p}_{13}], [\mathfrak{p}_{17}], [\mathfrak{p}'_{17}]$ and $[\mathfrak{p}''_{17}]$.

(We shall denote the identity element of $\text{Cl}(K_3)$ by $\mathbb{1}$.) We have the following relations

between the ideal classes of these generators:

(a) $\mathfrak{p}_2 \cdot \mathfrak{p}_2' \cdot \mathfrak{p}_2'' = (2) \sim (1) \Rightarrow [\mathfrak{p}_2] \cdot [\mathfrak{p}_2'] \cdot [\mathfrak{p}_2''] = \mathbb{1} \Rightarrow \mathfrak{p}_2'' \sim \mathfrak{p}_2^{-1} \cdot (\mathfrak{p}_2')^{-1}$

(b) $[\mathfrak{p}_7]^3 = \mathbb{1}$

(c) $[\mathfrak{p}_{13}]^3 = \mathbb{1}$

(d) $[\mathfrak{p}_{17}] \cdot [\mathfrak{p}_{17}'] \cdot [\mathfrak{p}_{17}''] = \mathbb{1}$

Therefore, $\mathrm{Cl}(K_3) = \langle\, [\mathfrak{p}_2], [\mathfrak{p}_2'], [\mathfrak{p}_7], [\mathfrak{p}_{13}], [\mathfrak{p}_{17}], [\mathfrak{p}_{17}'] \,\rangle$.

We are still left with the task of finding the relation between these generators. We make a few more norm computations:

We use **Sage** for computing the norms. For example, we can get the elements $\beta_3 \pm i$ in $\mathcal{O}_{K_3}$ for which the prime factorization of their norms $N_{K_3/\mathbb{Q}}(\beta_3 \pm i)$, for $1 \leq i < 100$ have a factor of 17 in them using the following code.

```
K.<b> = NumberField(x^3 -x^2- 30*x + 64)
l=[]
l2=[]
for i in range(1,100):
I2=K.fractional_ideal(b+i)
I3=K.fractional_ideal(b-i)
if 17.divides(int(I2.norm())) == True:
l.append((b+i, factor(I2.norm())))
l2.append(factor(I2.norm()))
if 17.divides(int(I3.norm())) == True:
l.append((b-i, factor(I3.norm())))
#print(l2)
l
```

We list some of these results below:

$$N_{K_3/\mathbb{Q}}(\beta_3 - 13) = 2 \cdot 23 \cdot 37 \qquad N_{K_3/\mathbb{Q}}(\beta_3 - 12) = 2^3 \cdot 7 \cdot 23$$

$$N_{K_3/\mathbb{Q}}(\beta_3 - 8) = 2^4 \cdot 17 \qquad N_{K_3/\mathbb{Q}}(\beta_3 - 11) = 2^4 \cdot 59$$

$$N_{K_3/\mathbb{Q}}(\beta_3 - 10) = 2^3 \cdot 83 \qquad N_{K_3/\mathbb{Q}}(\beta_3 - 9) = 2 \cdot 13 \cdot 17$$

$$N_{K_3/\mathbb{Q}}(\beta_3 - 5) = 2 \cdot 7 \qquad N_{K_3/\mathbb{Q}}(\beta_3 - 1) = 2 \cdot 17$$

$$N_{K_3/\mathbb{Q}}(\beta_3 + 2) = 7 \cdot 2^4$$

We observe the following: Without loss of generality, we fix that,

$$(\beta_3 - 1) = \mathfrak{p}_2 \cdot \mathfrak{p}_{17} \qquad (\beta_3 - 9) = \tilde{\mathfrak{p}_2} \cdot \mathfrak{p}_{13} \cdot \mathfrak{p}'_{17} \qquad (\beta_3 - 25) = \tilde{\mathfrak{p}_2} \cdot \mathfrak{p}''_{17} \cdot \mathfrak{p}_{421}$$

In the above, $\tilde{\mathfrak{p}_2}$ is used to denote that we are not aware which of $\mathfrak{p}_2, \mathfrak{p}'_2, \mathfrak{p}''_2$ occur in the factorization. We thus obtain that $[\mathfrak{p}_{13}] = \mathbb{1}$, $[\mathfrak{p}_{17}] = [\mathfrak{p}_2]^{-1}$ and $[\mathfrak{p}'_{17}] = [\tilde{\mathfrak{p}_2}]^{-1}$. Therefore, $\mathrm{Cl}(K) = \langle [\mathfrak{p}_2], [\mathfrak{p}'_2] \rangle$

We make a few more observations:

$$N_{K_3/\mathbb{Q}}(\beta_3^2 - 2\beta_3 + 1) = 2^2 \cdot 17^2 \qquad N_{K_3/\mathbb{Q}}(\beta_3^2 - 2\beta_3 - 1) = 2 \cdot 7$$
$$N_{K_3/\mathbb{Q}}(\beta_3^2 - 3\beta_3 - 3) = 7^2 \cdot 17 \qquad N_{K_3/\mathbb{Q}}(\beta_3^2 - 4\beta_3 + 2) = 2^2 \cdot 7 \cdot 17$$
$$N_{K_3/\mathbb{Q}}(\beta_3^2 - 3\beta_3 + 1) = 181 \qquad N_{K_3/\mathbb{Q}}(\beta_3^2 - 4\beta_3 - 2) = 2^2 \cdot 23$$

The following code now gives us elements of the form $\beta_3^2 + ib + j$ for some integers (which we can vary over a specified range) such that $4 \nmid N_{K_3/\mathbb{Q}}(\beta_3^2 + ib + j)$. This shall be useful as we get elements such that the maximum power of 2 dividing their norms is 1.

```
K.<b> = NumberField(x^3 -x^2- 30*x + 64)
l=[]
for i in range(-15,15):
  for j in range(-15,15):
    I2=K.fractional_ideal(b^2+i*b+j)
    if 4.divides(int(I2.norm())) == False:
      l.append((b^2+i*b+j, factor(I2.norm())))
l #l is a list of the elements we want whose norms are divisible by 2
   and not by 4
```

Using the above **Sage** code we make the following observations:

$$N_{K_3/\mathbb{Q}}(\beta_3^2 + 4\beta_3 - 15) = 2 \cdot 17 \qquad N_{K_3/\mathbb{Q}}(\beta_3^2 - 2\beta_3 - 1) = 2 \cdot 7$$
$$N_{K_3/\mathbb{Q}}(\beta_3^2 - 3\beta_3 - 3) = 7^2 \cdot 17 \qquad N_{K_3/\mathbb{Q}}(\beta_3^2 - 4\beta_3 + 2) = 2^2 \cdot 7 \cdot 17$$
$$N_{K_3/\mathbb{Q}}(\beta_3^2 - 3\beta_3 + 1) = 181 \qquad N_{K_3/\mathbb{Q}}(\beta_3^2 - 4\beta_3 - 2) = 2^2 \cdot 23$$

Ww find that from the norms which divisible by 2, we are unable to find a relation between the two generators $[\mathfrak{p}_2]$ and $[\mathfrak{p}'_2]$. We however expect the class group $\mathrm{Cl}(K_3)$ to be cyclic of order 3. We verify using **Sage** that the latter is in fact, true. △

Remark 5.4.3. Using similar techniques employed for Sub-extension 3, we find the 4th sub-extension of order 3 contained in $\mathbb{Q}(\zeta_{91})$ to be the splitting field of the polynomial $X^3 - X^2 - 30X - 27$ over $\mathbb{Q}$.

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